

MLib Yield Curve & Swaps Workshop

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MHI

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Introducing MLibrary

MLib Analytics Framework

- MHI use the MLib C++ Analytics Library
- 2 Excel Add-Ins Available: Vanilla-Addin & eTrading-Addin
- The eTrading-Addin is used for Swaps eTrading

Yield Curves:

- MLib Supports Multi-Yield Curves
- OIS Discounting and CSA Support is also Available
- In Excel a full set of EUR curves build in approx 0.5 second
i.e. EUROIS, EUR1ML, EUR3ML, EUR6ML and EUR12ML curves

Launcher

X:/Secure/DIR_MLib_Analytics/MLIB Excel Addin/LaunchETradingAddin.bat

Features

- Function Groups for Ease of Navigation: meCurve, meProducts ...
- Multi-Curve Construction and OIS Discounting with CSA Support
- OIS, IRS, Asset Swap Pricing & Risk
- Every Function has an Excel Example Template

Documentation

X:/Secure/DIR_MLib_Analytics/Documentation

Excel Examples

X:/Secure/DIR_MLib_Analytics/MLIB Examples

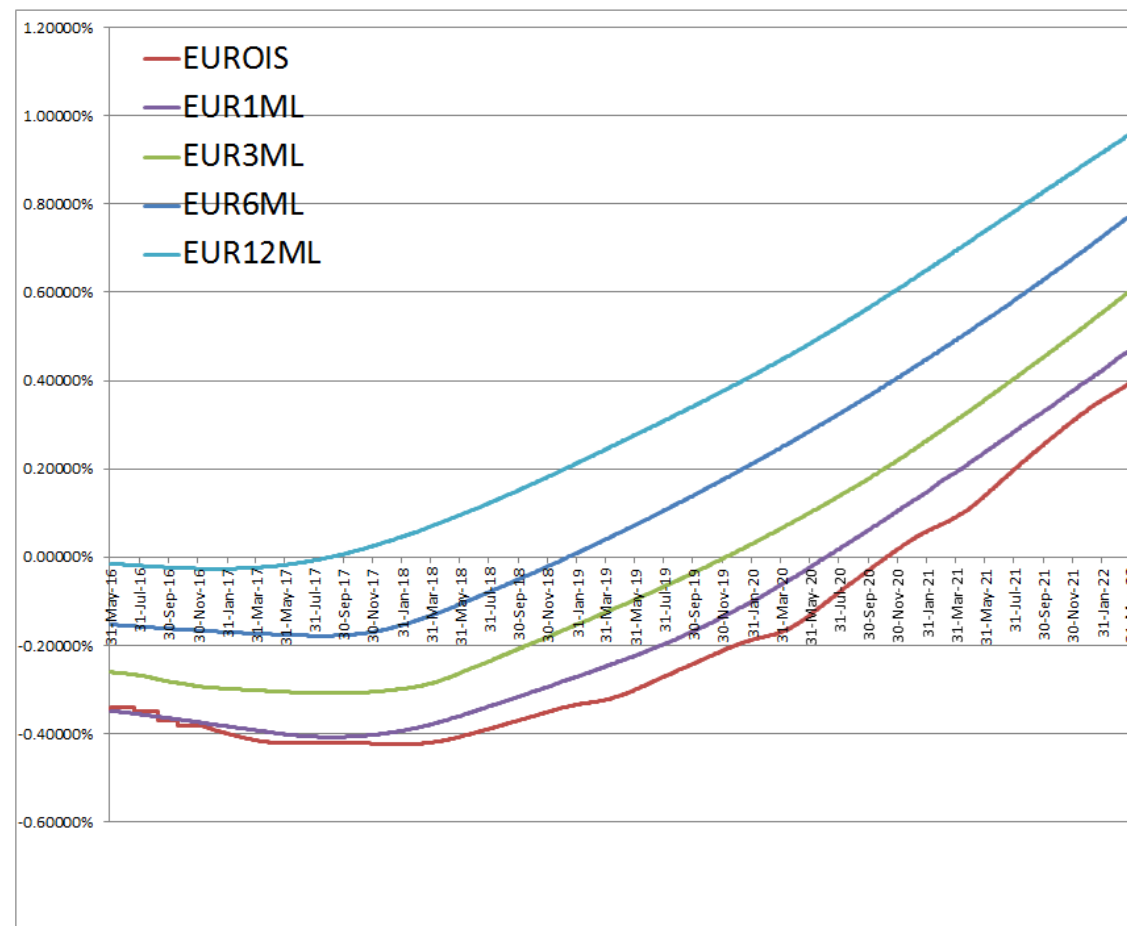
This Workshop

X:/Secure/DIR_MLib_Analytics/Workshop

Part I: Yield Curve Construction

Understanding & Configuring Curves

- Multiple curves needed for Libor forecast rates of different frequencies
- Tenor basis spreads are non-negligible
- Accounts for credit risk
- Calibration instruments must be tenor homogenous



Multi-Curves

MLib supports a full variety of curves including OIS, Standard Swap, Tenor Basis, Xccy Basis and CSA curves. These can be constructed using the below functions within the eTrading Excel addin

Curve Types:

`meCurveCalibrateOIS`

OIS Curves

`meCurveCalibrateSwap`

Standard Swap Curves

`meCurveCalibrateBasis`

Tenor Basis Curves

CSA Curves:

`meCurveCalibrateBasis`

Xccy Basis Curves

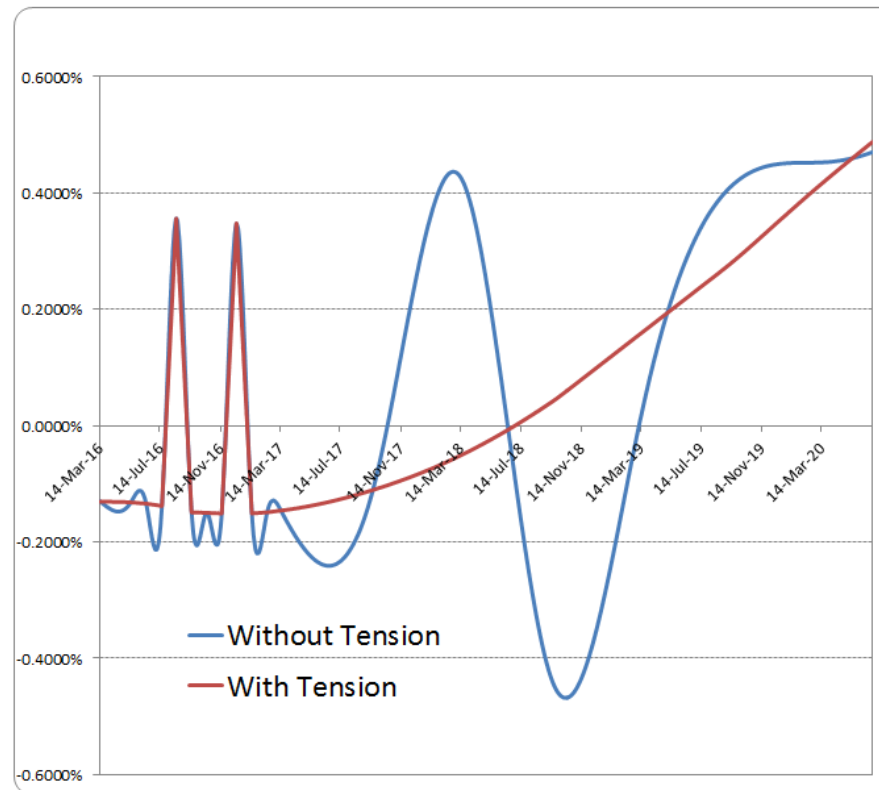
`meCurveCalibrateFXForwards`

CSA Curves

Current interpolation methods for curve construction:

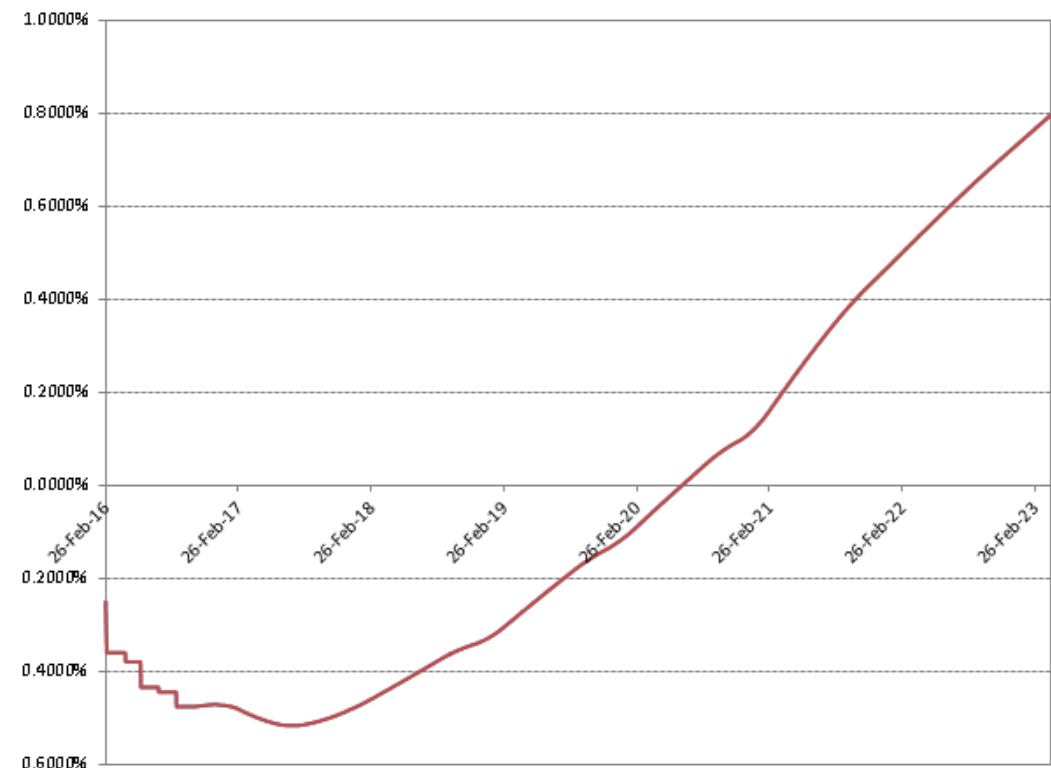
- Piecewise Constant `fn_stepInterpolation`
- Linear `fn_linearInterpolation`
- Monotone Convex `fn_monotoneconvexInterpolation`
- Spline `fn_splineInterpolation`
- Constrained Spline `fn_constrainedsplineInterpolation`

- Curve tension can be applied to swap curves
- Can be added to FRAs and / or Futures Instruments
- Used to manage curve over-oscillation & market data shocks



Calibration Instruments

- OIS Swaps
- Libor-OIS Basis Swaps
- Central Bank Dated Swaps
- e.g. ECB and Fed Fund Swaps



Curve Features

OIS curves are typically constructed using OIS Swaps and Libor-OIS Basis Swaps. The short end is interpolated piecewise constant between Central Bank meeting dates.

meCurveCalibrateOIS ("CurveCollection","StaticDataTable","CurveIndex","CurveBuildProperties","OISConventions","OISMarketData","OISFixingTable","OISBasisConventions","OISBasisMarketData","SwapConventions","SwapMarketData")

meCurveCalibrateOIS

meCurveCalibrateOIS EURYC OIS Curve has been set (0.02900 sec)	
Curve Loaded TRUE	
Curve Names	
CurveCollection	EURYC
StaticDataTable	OIS
IndexName(s)	OIS:EUROIS:EURDF:EURDF_NOCSA

CurveCollection: Name of the Curve Group, usually grouped by Currency
StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve parameters if not specified
IndexName: A user defined name(s) for the output curve index, mutiple aliases allowed. e.g. EUROIS, EUR3ML, EUR6ML

Curve Build Properties

Curve Build Properties	
AsOfDate	31-May-16
Currency	EUR
interpolation	fn_SplineInterpolation
yieldgen.interpolation	fn_SplineInterpolation

AsOfDate: The curve build Date
Currency: The curve currency
Interpolation: The interpolation method used for getting the OIS discount factors
YieldGen.Interpolation: The interpolation method used for generating the target curve forecast rates

OIS Curve Conventions

Overnight Indexed Swaps

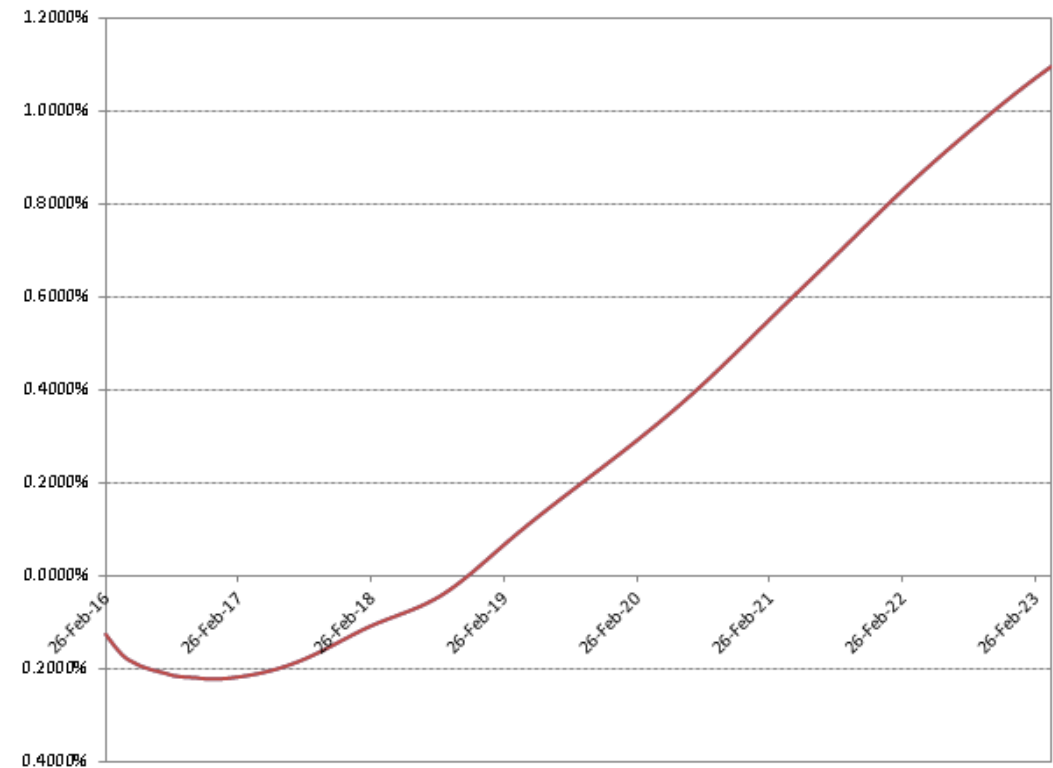
	OIS Conventions			OIS Par Rates				
				Term	Rate	StartDate	EndDate	Use
Calendar: The fixed leg calendar	calendar	TGT						
Daycount: The fixed leg daycount	daycount	ACT/360	CASH	ON	-0.345%			TRUE
Frequency: The fixed leg frequency	frequency	Annual	ECB SWAP	EUSF1A	-0.340%	02-Jun-16	21-Jul-16	TRUE
ResetLag: The float fixing lag	resetlag	2d	ECB SWAP	EUSF2A	-0.350%	21-Jul-16	08-Sep-16	TRUE
SlidingRule: The fixed leg BusinessDayAdjustment	slidingrule	MOD_FOLLOWING	ECB SWAP	EUSF3A	-0.371%	08-Sep-16	20-Oct-16	TRUE
IsEomRoll: Use the End of Month Roll Convention for swaps traded on End of Month	iseomroll	TRUE	ECB SWAP	EUSF4A	-0.382%	20-Oct-16	08-Dec-16	TRUE
EomDay: Day to be considered month end	eomday	31	OIS SWAP	12M	-0.382%			TRUE
GenerateMethod: Compounding method, can be "DailyCompounding" or remove this key and value for no compounding	generatemethod	dailycompounding	OIS SWAP	18M	-0.395%			TRUE
FirstRate: This field controls the first rate to be used for curve construction; "ShortMarket" implies include short end cash instruments. Removing this key and value will start curve construction from the Swaps instruments	firstrate	ShortMarket	OIS SWAP	2Y	-0.401%			TRUE
ShortTermConvention: The OIS compound method to be used. Can be "ArithmeticAverage" and if the key and value are removed will default to GeometricAverage	shorttermconvention	arithmeticaverage	OIS SWAP	3Y	-0.385%			TRUE
LongTermConvention: A flag to indicate if Libor OIS Spread instruments should be used on the Long End of the OIS Curve; Can be "LOBASIS" to use Libor OIS Spread instruments and if the key and value are removed only OIS Swap Outrights will	longtermconvention	LOBASIS	OIS SWAP	4Y	-0.342%			TRUE
LongTerm: The starting tenor (inclusive) for Libor-OIS Spread Instruments	longterm	3Y	OIS SWAP	5Y	-0.271%			TRUE
LongTerm.GenerateMethod: The compounding method for Libor OIS Spread instruments. Can be "DailyAveraging" or remove this key and value for no compounding.	longterm.generatemethod	dailyaveraging	OIS SWAP	6Y	-0.176%			TRUE
Epsilon: The solver precision to be used when calibrating the	Epsilon	1.00E-10	OIS SWAP	7Y	-0.066%			TRUE
MaxLoop: The maximum number of iterations to allow when calibrating the curve	MaxLoop	1,000	OIS SWAP	8Y	0.055%			TRUE
SmoothShortEnd: This flag controls if instruments before the Central Bank Swaps such as O/N and T/N rates should be included in the calibration routine; True = Include instruments such as O/N and T/N before Central Bank Swaps and false will exclude such instruments.	SmoothShortEnd	TRUE	OIS SWAP	9Y	0.175%			TRUE
			OIS SWAP	10Y	0.284%			TRUE

Libor-OIS Basis Swaps

	Libor-OIS Tenor Basis Conventions		Libor-OIS Spreads		Libor Swap Conventions		Libor Swap Rates	
Calendar: The fixed leg calendar Daycount: The fixed leg daycount Frequency: The fixed leg frequency SlidingRule: The fixed leg BusinessDayAdjustment	Calendar	TGT	Term	Rate	calendar	TGT	Term	Rate
	Daycount	ACT/360	2Y	0.111%	daycountfix	30/360	2Y	-0.293%
	Frequency	Quarterly	3Y	0.118%	frequencyfix	Annual	3Y	-0.271%
	SlidingRule	MOD_FOLLOWING	4Y	0.124%	slidingrule	MOD_FOLLOWING	4Y	-0.222%
			5Y	0.126%			5Y	-0.148%
			6Y	0.126%			6Y	-0.051%
			7Y	0.125%			7Y	0.060%
			8Y	0.122%			8Y	0.180%
			9Y	0.119%			9Y	0.298%
			10Y	0.118%			10Y	0.408%
			11Y	0.116%			11Y	0.510%
			12Y	0.115%			12Y	0.601%
			15Y	0.108%			15Y	0.807%
			20Y	0.096%			20Y	0.981%
			25Y	0.086%			25Y	1.037%
			30Y	0.077%			30Y	1.053%
			40Y	0.064%			40Y	1.053%
			50Y	0.056%			50Y	1.003%

Calibration Instruments

- Libor Rates
- FRAs
- Futures
- Swaps



Curve Features

Standard swap curves are using the above instruments. Curves are built using global optimization techniques having an analytical risk Jacobian. Furthermore Tension features are available

```
meCurveCalibrateSwap("CurveCollection","StaticDataTable","CurveIndex","CurveBuildProperties","MoneyMarketConventions",
LiborConventions,"LiborFixingTable","SwapConventions","SwapMarketData","FRAConventions","FRAMarketData3M",
"FRAMarketData6M","FuturesConventions","FuturesRates","ConvexityAdjustmentConventions","ConvexityAdjustmentMarketData")
```

meCurveCalibrateSwap

meCurveCalibrateSwap		EURYC STD Curve has been set (0.00800 sec)	
Curve Loaded		TRUE	

CurveCollection: Name of the Curve Group, usually grouped by Currency	
StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve parameters if not specified	
IndexName: A user defined name(s) for the output curve index, mutliple aliases allowed. e.g. EUROIS, EUR3ML, EUR6ML	

Curve Names	
CurveCollection	EURYC
StaticDataTable	STD
IndexName(s)	STD:EUR3ML

Curve Build Properties

AsOfDate:

The curve build Date

Currency:

The curve currency

Interpolation:

The interpolation method used for getting the OIS discount factors

YieldGen.Interpolation:

The interpolation method used for generating the target curve forecast rates

isFraUse:

Should FRA instruments be included in the calibration routine; TRUE = Include FRAs, FALSE = Exclude FRAs

isFutureUse:

Should Futures instruments be included in the calibration routine; TRUE = Include Futures, FALSE = Exclude Futures

Frequency:

The curve frequency, this controls which Libor Fixing should be associated with this curve.

dfCurveName:

Here we specify which discount factor curve should be used. Can be "OIS" or if left-blank Libor discounting will be assumed. (This field must not be used on the OIS Curve)

Curve Build Properties

AsOfDate	31-May-16
Currency	EUR
interpolation	fn_SplineInterpolation
yieldgen.interpolation	fn_SplineInterpolation
isfrause	FALSE
isfutureuse	TRUE
Frequency	Quarterly
dfcurvename	OIS

Cash Instruments

	Money Market O/N and T/N Conventions		Libor Fixing Conventions		Libor Fixing Data	
Calendar: The fixed leg calendar	Calendar	TGT	Calendar	TGT	Term	Rate
Daycount: The fixed leg daycount	DayCount	ACT/360	Daycount	ACT/360	1M	-0.34900%
SlidingRule: The fixed leg BusinessDayAdjustment	Slidingrule	FOLLOWING	Frequency	Simple	3M	-0.26100%
			Resetlag	2	6M	-0.15300%
			SlidingRule	MOD_FOLLOWING	12M	-0.01500%

* These fixings are used in the tenor basis curves see 'SpotRateTerm'

Futures Instruments

Calendar: The fixed leg calendar	Futures Conventions		Futures Data					
Daycount: The fixed leg daycount	calendar	TGT		Term / Ticker	StartDate	EndDate	Rate	Volatility
ResetLag: The float fixing lag	daycount	Act/360	4	ERU6 Comdty	21-Sep-16	21-Dec-16	100.2800	0.514%
SlidingRule: The fixed leg BusinessDayAdjustment	resetlag	2	5	ERV6 Comdty	19-Oct-16	18-Jan-17	100.2850	0.514%
MeanReversion: The mean reversion parameter for the convexity adjustment model. Use 0% for	slidingrule	MOD_FOLLOWING	6	ERX6 Comdty	16-Nov-16	15-Feb-17	100.2900	0.514%
ApplyTension: Insert sythetic points between	MeanReversion	3.00%	7	ERZ6 Comdty	21-Dec-16	15-Mar-17	100.2950	0.514%
TensionGap: The frequency in days for synthetic	ApplyTension	TRUE	8	ERH7 Comdty	15-Mar-17	21-Jun-17	100.3000	0.514%
	TensionGap	2	9	ERM7 Comdty	21-Jun-17	20-Sep-17	100.3050	0.514%
			10	ERU7 Comdty	20-Sep-17	20-Dec-17	100.3050	0.514%
			11	ERZ7 Comdty	20-Dec-17	21-Mar-18	100.3000	0.514%
			12	ERH8 Comdty	21-Mar-18	20-Jun-18	100.2850	0.514%

Swap Instruments

Calendar: The fixed leg calendar
Daycount: The fixed leg daycount
DaycountFloat: The float leg daycount
Frequency: The fixed leg frequency
FrequencyFloat: The float leg frequency
Interpolation: The interpolation method used for getting the OIS discount factors
IsEomRoll: Use the End of Month Roll Convention for swaps traded on End of Month
EomDay: Day to be considered month end
ResetLag: The float fixing lag
SlidingRule: The fixed leg BusinessDayAdjustment
IsNewtonRaphson: Use Newton Raphson solver for calibration. Currently the only valid option is "TRUE".
IsSimultaneousEq: Solve the Newton Raphson problem as a Simultaneous Equation. Currently the only valid option is "TRUE".
Epsilon: The solver precision to be used when calibrating the curve
MaxLoop: The maximum number of iterations to allow when calibrating the curve

Swap Conventions

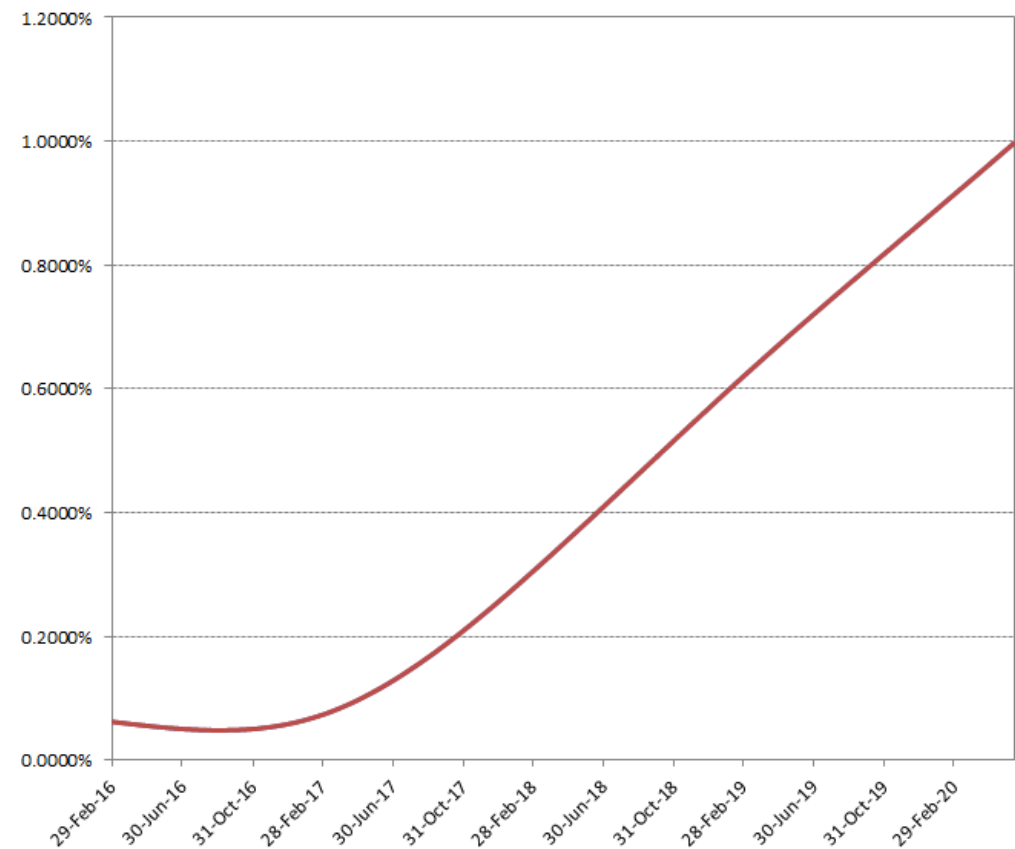
Calendar	TGT
DayCount	30/360
DayCountFloat	ACT/360
Frequency	Annual
FrequencyFloat	Quarterly
Interpolation	fn_SplineInterpolation
IsEomRoll	TRUE
EomDay	31
Resetlag	2
SlidingRule	MOD_FOLLOWING
IsNewtonraphson	TRUE
issimultaneouseq	TRUE
Epsilon	1.00E-09
MaxLoop	1,000

Swap Data

Term	Rate	Use
3Y	-0.26500%	TRUE
4Y	-0.20500%	TRUE
5Y	-0.12475%	TRUE
6Y	-0.02475%	TRUE
7Y	0.09025%	TRUE
8Y	0.21275%	TRUE
9Y	0.33225%	TRUE
10Y	0.44425%	TRUE
11Y	0.54333%	TRUE
12Y	0.63458%	TRUE
15Y	0.84325%	TRUE
20Y	1.01675%	TRUE
25Y	1.06967%	TRUE
30Y	1.08675%	TRUE

Calibration Instruments

- Tenor Basis Swaps
- or Xccy Basis Swaps



Curve Features

Both Tenor & Xccy Basis Curves are built using bootstrapping / solver techniques

```
meCurveCalibrateBasis("CurveCollection","StaticDataTable","CurveIndex","CurveBuildProperties","BasisSwapConventions",
"BasisSwapMarketData","FXForwardConventions","FXForwardMarketData","SpotFXMarketData")
```

meCurveCalibrateBasis

meCurveCalibrateBasis		EURYC 3M6MBASIS Curve has been set (0.00900 sec)	
Curve Loaded		TRUE	

Curve Names	
CurveCollection	EURYC
StaticDataTable	3M6MBASIS
IndexName(s)	3M6MBASIS:EUR6ML

CurveCollection: Name of the Curve Group, usually grouped by Currency
StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve parameters if not specified
IndexName: A user defined name(s) for the output curve index, mutiple aliases allowed. e.g. EUROIS, EUR3ML, EUR6ML

Curve Build Properties

AsOfDate: The curve build Date	Curve Build Properties
Currency: The curve currency	AsOfDate: 31-May-16
Basis.Interpolation The interpolation method used for generating the target curve i.e. interpolation method to use when calibrating this curve	Currency: EUR
isFwdBasis: Allow forward starting tenor basis; No = Don't allow forward starting Tenor Basis Spreads, True will allow the extra columns for forward starting instruments (highlighted in yellow) in the Tenor Basis Spreads table to be	basis.interpolation: fn_SplineInterpolation
dfCurveName: Here we specify which discount factor curve should be used. Can be "OIS" or if left-blank Libor discounting will be assumed. (This field must not be used on the OIS Curve)	isFwdBasis: TRUE
	dfCurveName: OIS

Tenor Basis Spread Instruments I

Tenor Basis Swaps

MarketName: StaticDataTable name, a hard-coded reference to a property file table used to lookup default curve parameters if not specified IsLeg1Spread: Should the Tenor Basis spread be added to Leg 1 of the Tenor Basis Swap? True = Add spread to Leg1, False = Add spread to Leg2 Leg1Cashlet.Calendar: Leg1 Accrual Calendar Leg1Cashlet.Daycount: Leg1 Accrual Daycount Leg1Cashlet.Frequency: Leg1 Accrual Frequency Leg1Cashlet.SlidingRule: Leg1 Accrual BusinessDayAdjustment or SlidingRule Leg1Cashlet.SpotLag: Leg1 Accrual Spot Lag (days) Leg1Discount: Leg1 Discount Curve e.g. EUROIS Leg1Forecast: Leg1 Forecast Curve e.g. EUR1ML, EUR3ML, EUR6ML, EUR12ML Leg1Index.Accessary: Leg1 Forecast Curve or Index Frequency Leg1Index.Daycount: Leg1 Payment Daycount Leg1Index.Fixing.Calendar: Leg1 Fixing Calendar Leg1Index.Frequency: Leg1 Yield Compound Frequency ; Simple, Continuous, Annual, Semi-Annual, Quarterly etc Leg1Index.Payment.Calendar: Leg1 Payment Calendar Leg1Index.ResetLag: Leg1 Fixing Lag (days) Leg1Index.SlidingRule: Leg1 Payment BusinessDayAdjustment or SlidingRule	Tenor Basis Conventions (3 X 6)		* Don't mix dates and tenors: Supply StartDate and EndDate or StartTenor and EndTenor Tenor Basis Spreads (6 X 12)				
	MarketName	EUR3M6MBASIS	Term	Rate	isDate	StartDate or Tenor	EndDate or Tenor
IsLeg1Spread		TRUE	6M	0.13000%	TRUE	02-Jun-16	02-Dec-16
Leg1cashlet.calendar		TGT	6M	0.13000%	TRUE	02-Dec-16	02-Jun-17
Leg1cashlet.daycount		Act/360	6M	0.13000%	TRUE	02-Jun-17	04-Dec-17
Leg1cashlet.frequency		Quarterly	6M	0.13000%	TRUE	04-Dec-17	04-Jun-18
Leg1cashlet.slidingrule		MOD_FOLLOWING	3Y	0.13000%	FALSE	0y	3Y
Leg1cashlet.spotlag		2	4Y	0.13500%	FALSE	0y	4Y
Leg1discount		OIS	5Y	0.13900%	FALSE	0y	5Y
Leg1forecast		EUR3ML	6Y	0.13900%	FALSE	0y	6Y
Leg1index.accessary		3M	7Y	0.13700%	FALSE	0y	7Y
Leg1index.daycount		Act/360	8Y	0.13250%	FALSE	0y	8Y
Leg1index.fixing.calendar		TGT	9Y	0.12600%	FALSE	0y	9Y
Leg1index.frequency		Simple	10Y	0.11900%	FALSE	0y	10Y
Leg1index.payment.calen		TGT	11Y	0.11250%	FALSE	0y	11Y
Leg1index.resetlag		2	12Y	0.10650%	FALSE	0y	12Y
Leg1index.slidingrule		MOD_FOLLOWING	15Y	0.09100%	FALSE	0y	15Y

Tenor Basis Spread Instruments II

Leg2Cashlet.Calendar: Leg2 Accrual Calendar	Leg2cashlet.calendar	TGT	20Y	0.07100%	FALSE	0y	20Y
Leg2Cashlet.Daycount: Leg2 Accrual Daycount	Leg2cashlet.daycount	Act/360	25Y	0.05825%	FALSE	0y	25Y
Leg2Cashlet.Frequency: Leg2 Accrual Frequency	Leg2cashlet.frequency	Semi-Annual	30Y	0.04900%	FALSE	0y	30Y
Leg2Cashlet.SlidingRule: Leg2 Accrual BusinessDayAdjustment or SlidingRule	Leg2cashlet.slidingrule	MOD_FOLLOWING					
Leg2Cashlet.SpotLag: Leg2 Accrual Spot Lag (days)	Leg2cashlet.spotlag	2					
Leg2Discount: Leg2 Discount Curve e.g. EUROIS	Leg2discount	OIS					
Leg2Forecast: Leg2 Forecast Curve e.g. EUR1ML, EUR3ML, EUR6ML, EUR12ML	Leg2forecast	STD	<--- Solver: Initial Guess				
Leg2Index.Accessary: Leg2 Forecast Curve or Index Frequency	Leg2index.accessary	6M					
Leg2Index.Daycount: Leg2 Payment Daycount	Leg2index.daycount	Act/360					
Leg2Index.Fixing.Calendar: Leg2 Fixing Calendar	Leg2index.fixing.calendar	TGT					
Leg2Index.Frequency: Leg2 Yield Compound Frequency; Simple, Continuous, Annual, Semi-Annual, Quarterly etc	Leg2index.frequency	Simple					
Leg2Index.Payment.Calendar: Leg2 Payment Calendar	Leg2index.payment.calen	TGT					
Leg2Index.ResetLag: Leg2 Fixing Lag (days)	Leg2index.resetlag	2					
Leg2Index.SlidingRule: Leg2 Payment BusinessDayAdjustment or SlidingRule	Leg2index.slidingrule	MOD_FOLLOWING					

Tenor Basis Spread Instruments III

Target:

What do we want to solve for

IsSimultaneousEq:

Calibrate and optimize using a Simultaneous Equation method. Currently the only valid option is "TRUE".

FwdInterpolation:

Interpolation method for interpolating Libor forecast rates

IsFwdInter:

Imply Libor Forecast rates by interpolating forwards; True = Imply forecast rates from Forwards, False = Imply forecast rates from Pseudo-Discount Factors

IsSameGridIndex:

Are the index fixing dates the same as the cashflow accrual dates i.e are the date grids the same? Recommended setting is "TRUE", sometimes when set to false curves will fail to solve and converge.

SpotRateTerm:

Tenor Basis curve fixings must be input in the Swap or STD curve Libor Fixing table. This field specifies which Libor fixing should be used when building the Tenor Basis Curve. Select 3M if building a 3M curve and 6M if building a 6M curve etc.

isYieldSpreadCalc:

Is the calibration target a spread on a base curve. This field should typically be set to "FALSE"

Epsilon:

The solver precision to be used when calibrating the curve

MaxLoop:

The maximum number of iterations to allow when calibrating the curve

Target

issimultaneouseq

fwdinterpolation

isfwdinter

issamegridindex

spotrateterm

isyieldspreadcalc

Epsilon

MaxLoop

Leg2Forecast

<--- Solver: Target

TRUE

fn_SplineInterpolation

TRUE

TRUE

6M

FALSE

1.00E-09

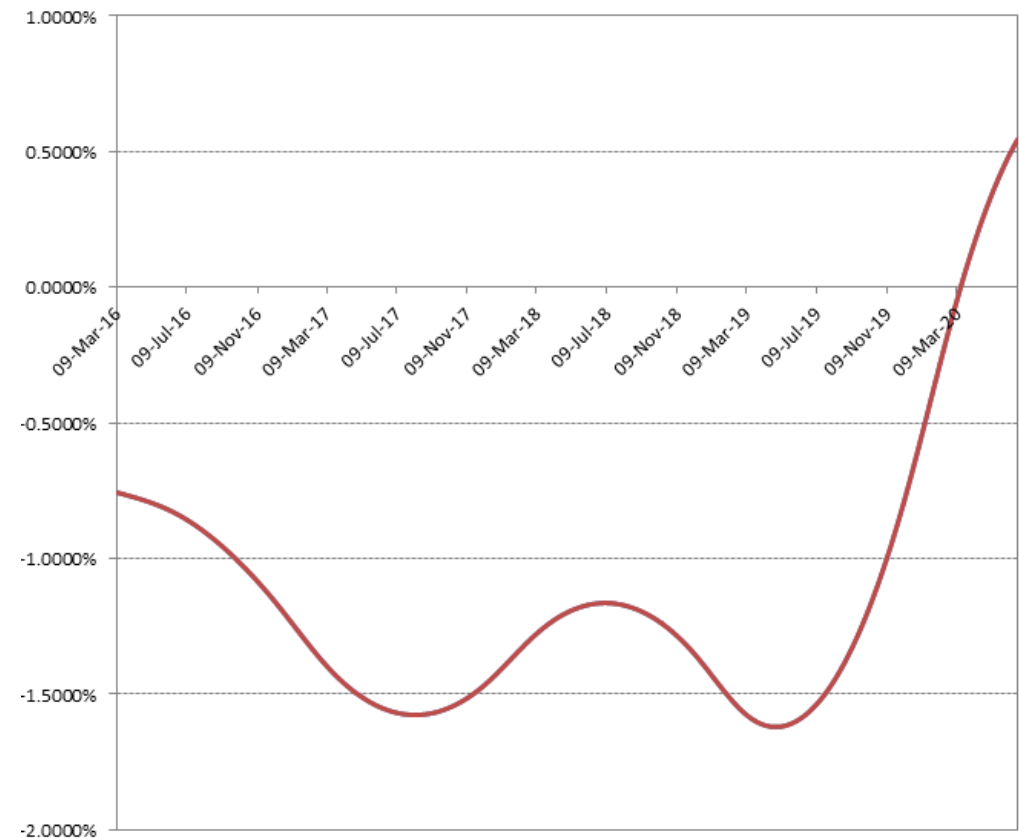
1,000

Build Method

- Bootstrapping
- Assumes constant FX Forwards or **FX Forward Invariance**

Calibration Instruments

- Xccy Basis Swaps
- FX Forwards



Curve Features

CSA curves are bootstrapped from Xccy swaps and assume FX Forward invariance, see next slide.

- We assume that FX Forwards are constant
- FX Forwards are fixed assuming a USD CSA

FX Forward Invariance - Using EUR/GBP as an Example

$$FwdFX(S, T)^{EUR/GBP} = s \frac{P(t, T)^{EUR-USDCSA}}{P(t, T)^{GBP-USDCSA}} = s \frac{P(t, T)^{EUR-GBPCSA}}{P(t, T)^{GBP-GBPCSA}}$$

where t denotes the valuation date, s the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t \leq S \leq T$

- From which Discounted CSA Curves can then be implied

Discount CSA Curve - General Formula

$$P(t, T)^{Ccy1-Ccy2CSA} = \left(\frac{P(t, T)^{Ccy1-USDCSA}}{P(t, T)^{Ccy2-USDCSA}} \right) \times \underbrace{P(t, T)^{Ccy2-Ccy2CSA}}_{\text{Ccy2 OIS Discount Factor}}$$

Build Order Native CSA Curves

- ① OIS Curve
- ② Standard Swap Curve(s)
- ③ Tenor Basis Curve(s)

OIS Discounting & CSA Curves

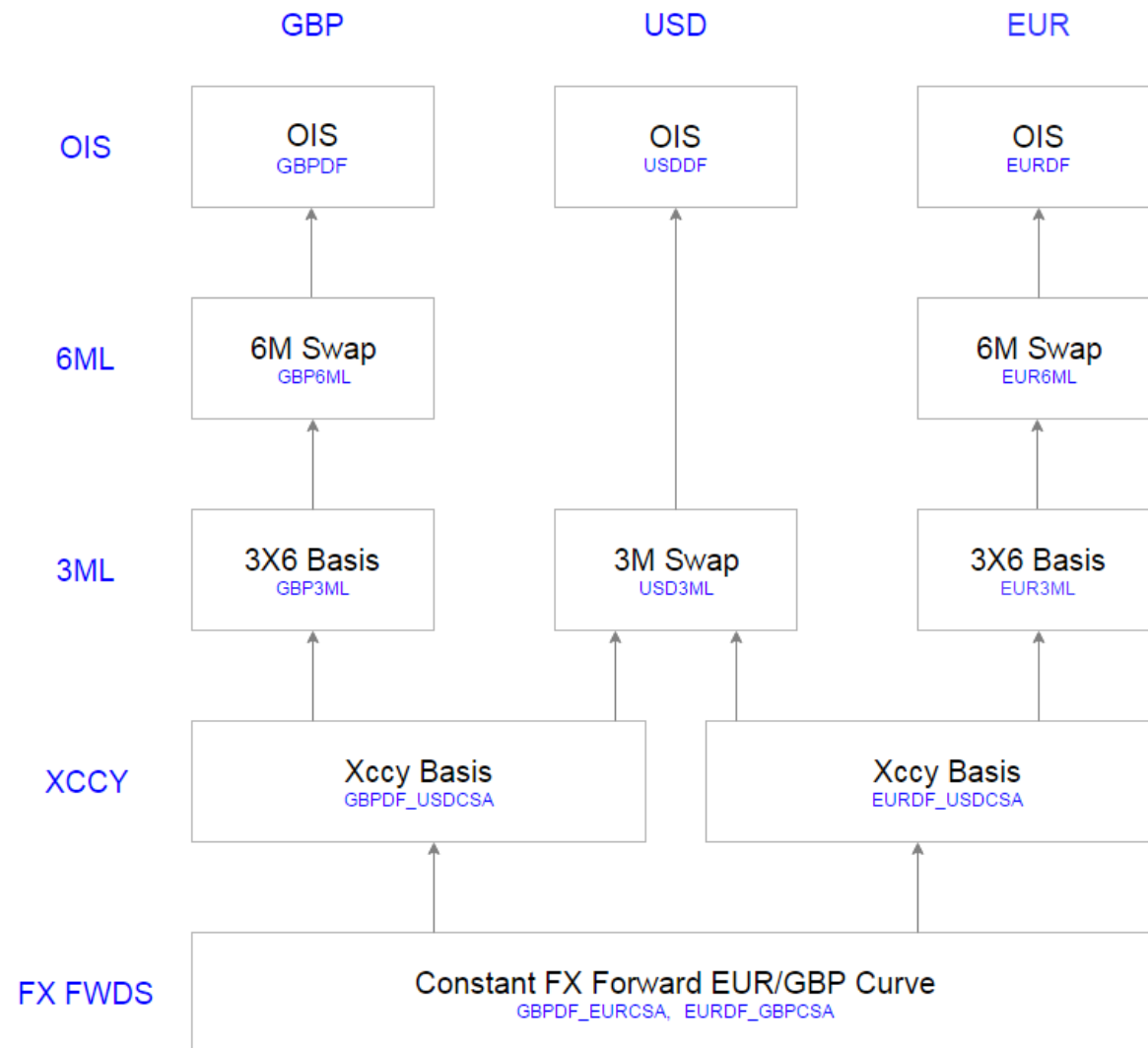
USD

	Instruments Used	USDYC
OIS	FED Funds vs 12m Fixed	OIS OIS Curve has been set (0.06200 sec)
STD	3m Float vs 6m Fixed	STD Swap Curve has been set (0.00000 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.01500 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 3x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

EUR

	Instruments Used	EURYC
OIS	EONIA vs 12m Fixed	OIS OIS Curve has been set (0.03100 sec)
STD	6m Float vs 12m Fixed	STD Swap Curve has been set (0.01500 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.00000 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 6x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

Non-Native CSA Curves have Additional Currency Dependencies



Swap Pricing, Native CSA

EUR Swaps

PV

-312,926

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		NOCSA	-0.13969%	-11,440	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		NOCSA	-0.13300%	-22,791	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		NOCSA	-0.08700%	-32,910	303
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		NOCSA	-0.00800%	-40,682	404
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		NOCSA	0.09300%	-45,732	504
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		NOCSA	0.20200%	-48,247	605
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		NOCSA	0.32100%	-47,815	704
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		NOCSA	0.44400%	-44,630	803
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		NOCSA	0.56100%	-39,518	900
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		NOCSA	0.66700%	-33,156	996
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		NOCSA	0.76300%	-25,831	1,090
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		NOCSA	0.84700%	-18,094	1,183
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		NOCSA	0.91775%	-10,476	1,274
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		NOCSA	0.97836%	-2,951	1,364
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		NOCSA	1.03000%	4,355	1,452
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		NOCSA	1.06960%	10,704	1,538
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		NOCSA	1.10207%	16,568	1,623
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		NOCSA	1.12836%	21,911	1,707
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		NOCSA	1.14941%	26,740	1,790
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		NOCSA	1.16600%	31,069	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	0.000
3Y	-0.08700%	0.000
4Y	-0.00800%	0.000
5Y	0.09300%	0.000
6Y	0.20200%	0.000
7Y	0.32100%	0.000
8Y	0.44400%	0.000
9Y	0.56100%	0.000
10Y	0.66700%	0.000
11Y	0.76300%	0.000
12Y	0.84700%	0.000
15Y	1.03000%	0.000
20Y	1.16600%	0.000
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-

Swap Pricing, Non-Native CSA

EUR Swaps

PV

-311,136

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		GBPCSA	-0.13970%	-11,442	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		GBPCSA	-0.13305%	-22,770	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		GBPCSA	-0.08711%	-32,855	302
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		GBPCSA	-0.00825%	-40,589	403
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		GBPCSA	0.09252%	-45,606	503
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		GBPCSA	0.20131%	-48,105	602
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		GBPCSA	0.32012%	-47,675	701
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		GBPCSA	0.44295%	-44,513	799
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		GBPCSA	0.55990%	-39,433	896
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		GBPCSA	0.66600%	-33,103	991
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		GBPCSA	0.76222%	-25,802	1,085
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		GBPCSA	0.84647%	-18,081	1,178
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		GBPCSA	0.91748%	-10,470	1,269
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		GBPCSA	0.97836%	-2,941	1,359
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		GBPCSA	1.03030%	4,386	1,447
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		GBPCSA	1.07020%	10,769	1,534
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		GBPCSA	1.10295%	16,680	1,620
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		GBPCSA	1.12949%	22,079	1,705
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		GBPCSA	1.15076%	26,971	1,789
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		GBPCSA	1.16753%	31,367	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	-0.005
3Y	-0.08700%	-0.011
4Y	-0.00800%	-0.025
5Y	0.09300%	-0.048
6Y	0.20200%	-0.069
7Y	0.32100%	-0.088
8Y	0.44400%	-0.105
9Y	0.56100%	-0.110
10Y	0.66700%	-0.100
11Y	0.76300%	-0.078
12Y	0.84700%	-0.053
15Y	1.03000%	0.030
20Y	1.16600%	0.153
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-
		-

Interest Rate Swaps, Eur3mL

EUR3ML

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate
1	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	3ML	0.00	NOCSA	-0.26500%
2	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	3ML	0.00	NOCSA	-0.20500%
3	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	3ML	0.00	NOCSA	-0.12475%
4	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	3ML	0.00	NOCSA	-0.02475%
5	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	3ML	0.00	NOCSA	0.09025%
6	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	3ML	0.00	NOCSA	0.21275%
7	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	3ML	0.00	NOCSA	0.33225%
8	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	3ML	0.00	NOCSA	0.44425%
9	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	3ML	0.00	NOCSA	0.54333%
10	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	3ML	0.00	NOCSA	0.63458%
11	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	3ML	0.00	NOCSA	0.84325%
12	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	3ML	0.00	NOCSA	1.01675%
13	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	25Y	1.00000%	3ML	0.00	NOCSA	1.06967%
14	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	30Y	1.00000%	3ML	0.00	NOCSA	1.08675%

Swap Calibration, Eur3mL

Term	Rate	+/- Bps
3Y	-0.26500%	0.000
4Y	-0.20500%	0.000
5Y	-0.12475%	0.000
6Y	-0.02475%	0.000
7Y	0.09025%	0.000
8Y	0.21275%	0.000
9Y	0.33225%	0.000
10Y	0.44425%	0.000
11Y	0.54333%	0.000
12Y	0.63458%	0.000
15Y	0.84325%	0.000
20Y	1.01675%	0.000
25Y	1.06967%	0.000
30Y	1.08675%	0.000

Calibration Results: Tenor Basis 3 X 6

Tenor Basis 3 X 6, Eur6mL

EUR3ML + Tenor Basis Spread

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
1	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	3Y	0.00000%	3ML	13.00	NOCSA	-0.13339%	-4,037
2	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	4Y	0.00000%	3ML	13.50	NOCSA	-0.06820%	-2,754
3	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	5Y	0.00000%	3ML	13.90	NOCSA	0.01612%	814
4	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	6Y	0.00000%	3ML	13.90	NOCSA	0.11616%	7,043
5	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	7Y	0.00000%	3ML	13.70	NOCSA	0.22919%	16,198
6	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	8Y	0.00000%	3ML	13.25	NOCSA	0.34723%	28,006
7	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	9Y	0.00000%	3ML	12.60	NOCSA	0.46018%	41,632
8	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	10Y	0.00000%	3ML	11.90	NOCSA	0.56512%	56,613
9	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	11Y	0.00000%	3ML	11.25	NOCSA	0.65764%	72,175
10	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	12Y	0.00000%	3ML	10.65	NOCSA	0.74285%	88,526
11	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	15Y	0.00000%	3ML	9.10	NOCSA	0.93582%	137,145
12	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	20Y	0.00000%	3ML	7.10	NOCSA	1.08903%	206,268
13	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	25Y	0.00000%	3ML	5.83	NOCSA	1.12898%	259,068
14	Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	30Y	0.00000%	3ML	4.90	NOCSA	1.13665%	303,802

Tenor Basis 3 X 6 Calibration, Eur6mL

Term	3 X 6 Spread	Par Rate Difference	PV Difference
3Y	13.00	0.00000%	0
4Y	13.50	0.00000%	0
5Y	13.90	0.00000%	0
6Y	13.90	0.00000%	0
7Y	13.70	0.00000%	0
8Y	13.25	0.00000%	0
9Y	12.60	0.00000%	0
10Y	11.90	0.00000%	0
11Y	11.25	0.00000%	0
12Y	10.65	0.00000%	0
15Y	9.10	0.00000%	0
20Y	7.10	0.00000%	0
25Y	5.83	0.00000%	0
30Y	4.90	0.00000%	0

EUR6ML

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
1	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	3Y	0.00000%	6ML	0.00	NOCSA	-0.13339%	4,037
2	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	4Y	0.00000%	6ML	0.00	NOCSA	-0.06820%	2,754
3	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	5Y	0.00000%	6ML	0.00	NOCSA	0.01612%	-814
4	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	6Y	0.00000%	6ML	0.00	NOCSA	0.11616%	-7,043
5	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	7Y	0.00000%	6ML	0.00	NOCSA	0.22919%	-16,198
6	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	8Y	0.00000%	6ML	0.00	NOCSA	0.34723%	-28,006
7	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	9Y	0.00000%	6ML	0.00	NOCSA	0.46018%	-41,632
8	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	10Y	0.00000%	6ML	0.00	NOCSA	0.56512%	-56,613
9	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	11Y	0.00000%	6ML	0.00	NOCSA	0.65764%	-72,175
10	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	12Y	0.00000%	6ML	0.00	NOCSA	0.74285%	-88,526
11	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	15Y	0.00000%	6ML	0.00	NOCSA	0.93582%	-137,145
12	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	20Y	0.00000%	6ML	0.00	NOCSA	1.08903%	-206,268
13	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	25Y	0.00000%	6ML	0.00	NOCSA	1.12898%	-259,068
14	Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	30Y	0.00000%	6ML	0.00	NOCSA	1.13665%	-303,802

Calibration Results: Tenor Basis 1 X 3

Tenor Basis 1 X 3, Eur1mL

EUR1ML + Tenor Basis Spread

Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	1Y	0.00000%	1ML	9.00	NOCSA	-0.28587%	-2,870
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	2Y	0.00000%	1ML	9.20	NOCSA	-0.29629%	-5,978
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	3Y	0.00000%	1ML	9.40	NOCSA	-0.26500%	-8,021
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	4Y	0.00000%	1ML	9.90	NOCSA	-0.20500%	-8,278
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	5Y	0.00000%	1ML	9.85	NOCSA	-0.12475%	-6,302
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	6Y	0.00000%	1ML	9.90	NOCSA	-0.02475%	-1,500
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	7Y	0.00000%	1ML	9.90	NOCSA	0.09025%	6,379
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	8Y	0.00000%	1ML	9.20	NOCSA	0.21275%	17,160
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	9Y	0.00000%	1ML	9.20	NOCSA	0.33225%	30,059
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	10Y	0.00000%	1ML	8.90	NOCSA	0.44425%	44,505
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	15Y	0.00000%	1ML	7.35	NOCSA	0.84325%	123,579
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	20Y	0.00000%	1ML	5.35	NOCSA	1.01675%	192,578
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	25Y	0.00000%	1ML	4.08	NOCSA	1.06967%	245,459
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	30Y	0.00000%	1ML	3.15	NOCSA	1.08675%	290,467

Tenor Basis 1 X 3 Calibration, Eur1mL

Term	1 X 3 Spread	Par Rate Difference	PV Difference
1Y	9.00	0.00000%	0
2Y	9.20	0.00000%	0
3Y	9.40	0.00000%	0
4Y	9.90	0.00000%	0
5Y	9.85	0.00000%	0
6Y	9.90	0.00000%	0
7Y	9.90	0.00000%	0
8Y	9.20	0.00000%	0
9Y	9.20	0.00000%	0
10Y	8.90	0.00000%	0
15Y	7.35	0.00000%	0
20Y	5.35	0.00000%	1
25Y	4.08	0.00000%	1
30Y	3.15	0.00000%	1

EUR3ML

Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	1Y	0.00000%	3ML	0.00	NOCSA	-0.28587%	2,870
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	2Y	0.00000%	3ML	0.00	NOCSA	-0.29629%	5,978
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	3Y	0.00000%	3ML	0.00	NOCSA	-0.26500%	8,021
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	4Y	0.00000%	3ML	0.00	NOCSA	-0.20500%	8,278
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	5Y	0.00000%	3ML	0.00	NOCSA	-0.12475%	6,302
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	6Y	0.00000%	3ML	0.00	NOCSA	-0.02475%	1,501
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	7Y	0.00000%	3ML	0.00	NOCSA	0.09025%	-6,378
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	8Y	0.00000%	3ML	0.00	NOCSA	0.21275%	-17,160
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	9Y	0.00000%	3ML	0.00	NOCSA	0.33225%	-30,058
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	10Y	0.00000%	3ML	0.00	NOCSA	0.44425%	-44,505
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	15Y	0.00000%	3ML	0.00	NOCSA	0.84325%	-123,579
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	20Y	0.00000%	3ML	0.00	NOCSA	1.01675%	-192,578
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	25Y	0.00000%	3ML	0.00	NOCSA	1.06967%	-245,458
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	30Y	0.00000%	3ML	0.00	NOCSA	1.08675%	-290,466

Calibration Results: Tenor Basis 6 X 12

Tenor Basis 6 X 12

EUR3ML + Tenor Basis Spread

Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	2Y	0.00000%	6ML	15.00	NOCSA	-0.01652%	-333
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	3Y	0.00000%	6ML	15.30	NOCSA	0.02158%	653
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	4Y	0.00000%	6ML	15.30	NOCSA	0.08690%	3,509
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	5Y	0.00000%	6ML	15.00	NOCSA	0.16819%	8,497
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	6Y	0.00000%	6ML	14.70	NOCSA	0.26522%	16,080
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	7Y	0.00000%	6ML	14.10	NOCSA	0.37219%	26,305
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	8Y	0.00000%	6ML	13.40	NOCSA	0.48321%	38,975
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	9Y	0.00000%	6ML	12.80	NOCSA	0.59011%	53,387
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	10Y	0.00000%	6ML	12.20	NOCSA	0.68898%	69,022
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	15Y	0.00000%	6ML	9.30	NOCSA	1.03033%	150,996
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	20Y	0.00000%	6ML	7.80	NOCSA	1.16834%	221,290
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	25Y	0.00000%	6ML	7.00	NOCSA	1.20017%	275,404
Thu 2-Jun-16	2D	1,000,000	EUR	PAYER	30Y	0.00000%	6ML	6.40	NOCSA	1.20174%	321,200

Tenor Basis 6 X 12 Calibration, Eur12mL

Term	6 X 12 Spread	Par Rate Difference	PV Difference
2Y	15.00	0.00000%	0
3Y	15.30	0.00000%	0
4Y	15.30	0.00000%	0
5Y	15.00	0.00000%	0
6Y	14.70	0.00000%	0
7Y	14.10	0.00000%	0
8Y	13.40	0.00000%	0
9Y	12.80	0.00000%	0
10Y	12.20	0.00000%	0
15Y	9.30	0.00000%	0
20Y	7.80	0.00000%	0
25Y	7.00	0.00000%	0
30Y	6.40	0.00000%	0

EUR12ML

Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	CSA	Par Rate	Present Value
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	2Y	0.00000%	12ML	0.00	NOCSA	-0.01652%	333
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	3Y	0.00000%	12ML	0.00	NOCSA	0.02158%	-653
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	4Y	0.00000%	12ML	0.00	NOCSA	0.08690%	-3,509
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	5Y	0.00000%	12ML	0.00	NOCSA	0.16819%	-8,497
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	6Y	0.00000%	12ML	0.00	NOCSA	0.26522%	-16,080
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	7Y	0.00000%	12ML	0.00	NOCSA	0.37219%	-26,305
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	8Y	0.00000%	12ML	0.00	NOCSA	0.48321%	-38,975
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	9Y	0.00000%	12ML	0.00	NOCSA	0.59011%	-53,387
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	10Y	0.00000%	12ML	0.00	NOCSA	0.68898%	-69,022
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	15Y	0.00000%	12ML	0.00	NOCSA	1.03033%	-150,996
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	20Y	0.00000%	12ML	0.00	NOCSA	1.16834%	-221,290
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	25Y	0.00000%	12ML	0.00	NOCSA	1.20017%	-275,404
Thu 2-Jun-16	2D	1,000,000	EUR	RECEIVER	30Y	0.00000%	12ML	0.00	NOCSA	1.20174%	-321,200

End of Part I

MLib Yield Curve & Swaps Workshop

Nicholas Burgess



MHI

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14th June 2016

Part II: Swap Pricing & Risk

Swap Specification & Pricing

To specify a swap many parameters are required to generate the swap cashflow schedules accurately. To price a swap we require Libor forecast rates, OIS discount rates and a Swap pricing formula.

$$PV^{Swap} = N \sum_{\forall i} r^{Fixed} \tau_i P(t_0, t_i) - N \sum_{\forall j} (L_j + s) \tau_j P(t_0, t_j)$$



Fixed Leg

$$PV(Fixed) = N \times r^{Fixed} \underbrace{\sum_{\forall i} \tau_i P(t_0, t_i)}_{Annuity}$$

Float Leg

$$PV(Float) = N \sum_{\forall j} (L_j + s) \tau_j P(t_0, t_j)$$

Swap Price¹

$$PV(Swap) = \phi (PV(Fixed) - PV(Float))$$

Swap Rate

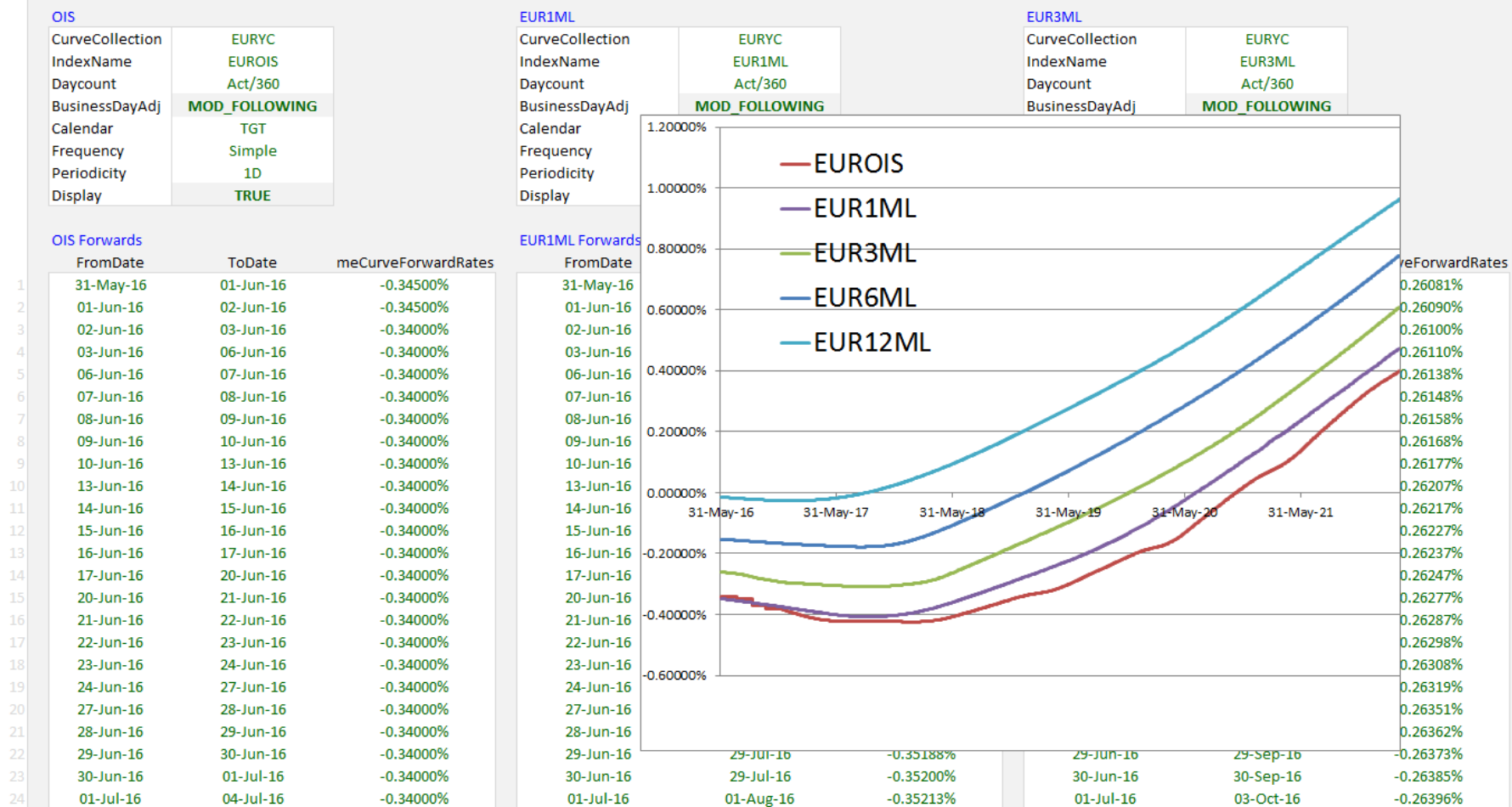
$$ParRate = \frac{PV(Float)}{N \times Annuity}$$

¹ $\phi=1$ if receiving and -1 if paying fixed coupons

- Forwards can be retrieved using `meCurveForwardRates`

`meCurveForwardRates("FromDate","ToDate","CurveCollection","CurveIndex","Daycount","BusinessDayAdj","Calendar","Frequency")`

`meCurveForwardRates`



- Discount Factors can be retrieved using `meCurveDiscountFactors`

meCurveDiscountFactorsFromYearFractions

DiscountFactor(s)	YearFraction
1.001828	0.500

CurveCollection	EURYC
CurveIndex	OIS

meCurveDiscountFactorsFromYearFractions (multiple inputs)

DiscountFactor(s)	YearFraction
1.001828	0.500
1.006036	1.500
1.008183	2.000
1.010127	2.500

CurveCollection	EURYC
CurveIndex	OIS

meCurveDiscountFactorsFromTenors

DiscountFactor(s)	Tenor
1.001828	6M

CurveID	EURYC
CurveIndex	OIS
BusinessDayAdjustment	MOD_FOLLOWING
Calendar	TGT

meCurveDiscountFactorsFromTenors (multiple inputs)

DiscountFactor(s)	Tenor
1.001828	6M
1.003884	1Y
1.006035	18M
1.008177	2Y

CurveID	EURYC
CurveIndex	OIS
BusinessDayAdjustment	MOD_FOLLOWING
Calendar	TGT

- Discount Factors can be retrieved using `meCurveDiscountFactors`

meCurveDiscountFactorsForwardStarting

DiscountFactor(s)	StartDate	EndDate
0.000000	19-Feb-16	19-Aug-16

meCurveDiscountFactorsForwardStarting (multiple inputs)

DiscountFactor(s)	StartDate	EndDate
1.001738	19-Feb-16	19-Aug-16
1.003758	19-Feb-16	19-Feb-17
1.005790	19-Feb-16	19-Aug-17
1.007687	19-Feb-16	19-Feb-18

meCurveDiscountFactorsForwardStartingFromYearFractions

DiscountFactor(s)	StartDate	YearFraction
1.000774	19-Feb-16	0.50

meCurveDiscountFactorsForwardStartingFromYearFractions (multiple inputs)

DiscountFactor(s)	StartDate	YearFraction
1.001748	19-Feb-16	0.50
1.001760	19-Mar-16	0.50
1.001813	19-Apr-16	0.50
1.001852	19-May-16	0.50

meCurveDiscountFactorsForwardStartingFromTenor

DiscountFactor(s)	StartDate	Term (tenor)
1.000764	19-Feb-16	6M

meCurveDiscountFactorsForwardStartingFromTenor (multiple inputs)

DiscountFactor(s)	StartDate	Term (tenor)
1.001738	19-Feb-16	6M
1.001771	19-Mar-16	6M
1.001813	19-Apr-16	6M
1.001884	19-May-16	6M

meProductSwapSchedule(ScheduleLVB)

Fixed Leg Specification

EffectiveDate	02-Jun-16
MaturityDate	5Y
Daycount	30/360
AccrualFrequency	Annual
AccrualBusinessDayAdjustment	MOD_FOLLOWING
AccrualCalendar	TGT
PaymentFrequency	Annual
PaymentBusinessDayAdjustment	MOD_FOLLOWING
PaymentCalendar	TGT
PaymentLag	0D
FixingBusinessDayAdjustment	MOD_FOLLOWING
FixingCalendar	TGT
FixingLag	0D
FixingAdvanceOrArrear	ADVANCE

Fixed Leg

	Fixing Date	Accrual Start	Accrual End	Accrual Period	Payment Date
0					
1	02-Jun-16	02-Jun-16	02-Jun-17	1.0000	02-Jun-17
2	02-Jun-17	02-Jun-17	04-Jun-18	1.0056	04-Jun-18
3	04-Jun-18	04-Jun-18	03-Jun-19	0.9972	03-Jun-19
4	03-Jun-19	03-Jun-19	02-Jun-20	0.9972	02-Jun-20
5	02-Jun-20	02-Jun-20	02-Jun-21	1.0000	02-Jun-21
6	#N/A	#N/A	#N/A	#N/A	#N/A
7	#N/A	#N/A	#N/A	#N/A	#N/A
8	#N/A	#N/A	#N/A	#N/A	#N/A
9	#N/A	#N/A	#N/A	#N/A	#N/A
10	#N/A	#N/A	#N/A	#N/A	#N/A
11	#N/A	#N/A	#N/A	#N/A	#N/A
12	#N/A	#N/A	#N/A	#N/A	#N/A
13	#N/A	#N/A	#N/A	#N/A	#N/A

Float Leg Specification

EffectiveDate	02-Jun-16
MaturityDate	5Y
Daycount	ACT/360
AccrualFrequency	Semi-Annual
AccrualBusinessDayAdjustment	MOD_FOLLOWING
AccrualCalendar	TGT
PaymentFrequency	Semi-Annual
PaymentBusinessDayAdjustment	MOD_FOLLOWING
PaymentCalendar	TGT
PaymentLag	0D
FixingBusinessDayAdjustment	MOD_FOLLOWING
FixingCalendar	TGT
FixingLag	0D
FixingAdvanceOrArrear	ADVANCE

Float Leg

	Fixing Date	Accrual Start	Accrual End	Accrual Period	Payment Date
0					
1	02-Jun-16	02-Jun-16	02-Dec-16	0.5083	02-Dec-16
2	02-Dec-16	02-Dec-16	02-Jun-17	0.5056	02-Jun-17
3	02-Jun-17	02-Jun-17	04-Dec-17	0.5139	04-Dec-17
4	04-Dec-17	04-Dec-17	04-Jun-18	0.5056	04-Jun-18
5	04-Jun-18	04-Jun-18	03-Dec-18	0.5056	03-Dec-18
6	03-Dec-18	03-Dec-18	03-Jun-19	0.5056	03-Jun-19
7	03-Jun-19	03-Jun-19	02-Dec-19	0.5056	02-Dec-19
8	02-Dec-19	02-Dec-19	02-Jun-20	0.5083	02-Jun-20
9	02-Jun-20	02-Jun-20	02-Dec-20	0.5083	02-Dec-20
10	02-Dec-20	02-Dec-20	02-Jun-21	0.5056	02-Jun-21
11	#N/A	#N/A	#N/A	#N/A	#N/A
12	#N/A	#N/A	#N/A	#N/A	#N/A
13	#N/A	#N/A	#N/A	#N/A	#N/A

Fixed Leg Cashflows

Notional	FixedRate	Coupon	DiscFact	CouponPV	
1,000,000	1.0000%	10,000	1.0039	10,039	-
1,000,000	1.0000%	10,056	1.0082	10,138	-
1,000,000	1.0000%	9,972	1.0118	10,090	-
1,000,000	1.0000%	9,972	1.0141	10,112	-
1,000,000	1.0000%	10,000	1.0139	10,139	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-
#N/A	#N/A	#N/A	#N/A	#N/A	-

Float Leg Cashflows

Notional	FloatRate	Spread	Coupon	DiscFact	CouponPV
1,000,000	-0.1530%	0	-778	1.0018	-779
1,000,000	-0.1666%	0	-842	1.0039	-846
1,000,000	-0.1763%	0	-906	1.0061	-912
1,000,000	-0.1691%	0	-855	1.0082	-862
1,000,000	-0.1050%	0	-531	1.0102	-536
1,000,000	-0.0200%	0	-101	1.0118	-102
1,000,000	0.0736%	0	372	1.0131	377
1,000,000	0.1758%	0	894	1.0141	906
1,000,000	0.2869%	0	1,458	1.0143	1,479
1,000,000	0.4076%	0	2,061	1.0139	2,089
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

Manual Swap Calculations

Swap Par Rate	0.0161%	PV Fixed	50,519
Swap PV	49,705	PV Float	814
PV01	505	Annuity	5,051,931

MLib Swap Deal Info

Notional	1,000,000
PayerReceiver	RECEIVER
EffectiveDate	Thu 2-Jun-16
MaturityDate	5Y
CurveCollection	EURYC
ForecastCurve	EUR6ML
DiscountCurve	OIS
FixedFrequency	Annual
FixedDaycount	30/360
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING
FixedPaymentCalendar	TGT
FixedRollDay	NORMAL
FixedPaymentLag	0D
FixedStubType	ShortStart
FixedRate	1.00%
FloatFrequency	Semi-Annual
FloatDaycount	ACT/360
FloatFixingBusinessDayAdjustment	MOD_FOLLOWING
FloatFixingCalendar	TGT
FloatAccrualBusinessDayAdjustment	MOD_FOLLOWING
FloatAccrualCalendar	TGT
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING
FloatPaymentCalendar	TGT
FloatFirstStubDate	
FloatLastStubDate	
FloatRollDay	NORMAL
FloatFixingLag	0D
FloatStubType	ShortStart
FloatSpread	0.00

MLib Swap Calculations

Swap Par Rate	0.0161%
Swap PV	49,705
Swap PV01	505
Swap DV01	521

Overnight Index Swap Pricing

meProductOISPV(DealInfoLVB)

meProductOISParRate(DealInfoLVB)

OIS Deal Info

Deal 1	
Notional	1,000,000
PayerReceiver	RECEIVER
EffectiveDate	02-Jun-16
CurveCollection	EURYC
ForecastCurve	OIS
DiscountCurve	EURDF_NOCSA
CompoundMethod	GEOMETRIC
MaturityDate	1Y
FixedFrequency	Annual
FixedDaycount	Act/360
FixedAccrualBusinessDayAdjustment	MOD_FOLLOWING
FixedAccrualCalendar	TGT
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING
FixedPaymentCalendar	TGT
FixedRollDay	NORMAL
FixedPaymentLag	0D
FixedStubType	ShortEnd
FixedRate	1.00%
FloatFrequency	Annual
FloatDaycount	Act/360
FloatFixingBusinessDayAdjustment	MOD_FOLLOWING
FloatFixingCalendar	TGT
FloatAccrualBusinessDayAdjustment	MOD_FOLLOWING
FloatAccrualCalendar	TGT
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING
FloatPaymentCalendar	TGT
FloatRollDay	NORMAL
FloatPaymentLag	0D
FloatStubType	ShortEnd
FloatSpread	0.00
FloatFixingLag	2D

MLIB OIS Price

PV	14,066.69
Par Rate	-0.3820%

3) Main		4) Details		5) Curves		6) Cashflow		7) Resets		9) Scenario		10) Risk		11) Valuation	
+ Deal		OIS Swap				Counterparty				SWAP CNTRPARTY					
- Swap		OIS Swap				OIS Swap				OIS Swap					
Leg 1:Fixed ▾		Receive ▾				Leg 2:Float ▾				Pay ▾					
Notional		1MM				Notional				1MM					
Currency		EUR ▾				Currency				EUR ▾					
Effective		0D	06/16/2016		Effective		0D	06/16/2016		Effective		0D	06/16/2016		
Maturity		1Y	06/16/2017		Maturity		1Y	06/16/2017		Maturity		1Y	06/16/2017		
Coupon		1.000000 %				Index				1D	EONIA				
Pay Freq		Annual ▾				Spread				0.000		bp			
Day Count		ACT/360 ▾				Leverage				1.00000					
Calc Basis		Money Mkt ▾				Latest Index				-0.32700					
						Reset Freq				Daily					
						Pay Freq				Annual ▾					
						Day Count				ACT/360 ▾					
+ Market															
- Leg 1: NPV		10,180.08				Leg 2: NPV				4,062.36					
Accrued		0.00				Accrued				0.00					
Premium		1.02				Premium				0.41					
DV01		1.04				DV01				102.21					
≡ Valuation Results															
Par Cpn		-0.399050				Premium				1.42424					
Principal		14,242.44				BP Value				142.42436					
Accrued		0.00				Average Coupon				0.39905					
NPV		14,242.44													

Bloomberg OIS Price

meProductSwapPV(DealInfoLVB)
meProductSwapParRate(DealInfoLVB)

meProductSwapPV01(DealInfoLVB)
meProductSwapDV01(DealInfoLVB)

Swap Deal Info

Deal 1	
Notional	1,000,000
PayerReceiver	RECEIVER
EffectiveDate	Thu 2-Jun-16
MaturityDate	5Y
CurveCollection	EURYC
ForecastCurve	EUR6ML
DiscountCurve	OIS
FixedFrequency	Annual
FixedDaycount	30/360
FixedAccrualBusinessDayAdjustment	MOD_FOLLOWING
FixedAccrualCalendar	TGT
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING
FixedPaymentCalendar	TGT
FixedRollDay	NORMAL
FixedPaymentLag	0D
FixedStubType	ShortStart
FixedRate	1.00%
FloatFrequency	Semi-Annual
FloatDaycount	ACT/360
FloatFixingBusinessDayAdjustment	MOD_FOLLOWING
FloatFixingCalendar	TGT
FloatAccrualBusinessDayAdjustment	MOD_FOLLOWING
FloatAccrualCalendar	TGT
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING
FloatPaymentCalendar	TGT
FloatRollDay	NORMAL
FloatPaymentLag	0D
FloatStubType	ShortStart
FloatSpread	0.00

Swap Pricing & Risk

Swap Par Rate	0.0161%
Swap PV	49,705
Swap PV01	505
Swap DV01	521

3) Main		4) Details		5) Curves		6) Cashflow		7) Resets		9) Scenario		10) Risk		11) ...			
Deal		Fixed Float Swap				Counterparty				SWAP CNTRPARTY							
Swap		Leg 1:Fixed ▾				Receive ▾				Leg 2:Float ▾				Pay ▾			
Notional						1MM				Notional				1MM			
Currency						EUR ▾				Currency				EUR ▾			
Effective		0D		06/16/2016				Effective		0D		06/16/2016					
Maturity		5Y		06/16/2021				Maturity		5Y		06/16/2021					
Coupon						1.000000 %				Index				6M EUR006M			
Pay Freq						Annual ▾				Spread				0.000 bp			
Day Count						30U/360 ▾				Leverage				1.00000			
Calc Basis						Money Mkt ▾				Latest Index				-0.15900			
Market										Reset Freq				SemiAnnual ▾			
										Pay Freq				SemiAnnual ▾			
										Day Count				ACT/360 ▾			
Leg 1: NPV						50,595.04				Leg 2: NPV				1,578.57			
Accrued						0.00				Accrued				0.00			
Premium						5.06				Premium				0.16			
DV01						15.48				DV01				506.44			
Valuation Results																	
Par Cpn						-0.031200				Premium		5.21736					
Principal						52,173.60				BP Value		521.73602					
Accrued						0.00											
NPV						52,173.60											

Bloomberg OIS Price

Tenor Basis Pricing

Swap Deal Info

	Leg1	Leg2
Notional	1,000,000	1,000,000
PayerReceiver	RECEIVER	PAYER
EffectiveDate	Thu 2-Jun-16	Thu 2-Jun-16
MaturityDate	5Y	5Y
CurveCollection	EURYC	EURYC
ForecastCurve	EUR3ML	EUR6ML
DiscountCurve	OIS	OIS
FixedFrequency	Annual	Annual
FixedDaycount	30/360	30/360
FixedAccrualBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FixedAccrualCalendar	TGT	TGT
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FixedPaymentCalendar	TGT	TGT
FixedFirstStubDate		
FixedLastStubDate		
FixedRollDay	NORMAL	NORMAL
FixedPaymentLag	0D	0D
FixedStubType	ShortStart	ShortStart
FixedRate	0.00%	0.00%
FloatFrequency	Quarterly	Semi-Annual
FloatDaycount	ACT/360	ACT/360
FloatFixingBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FloatFixingCalendar	TGT	TGT
FloatAccrualBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FloatAccrualCalendar	TGT	TGT
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
FloatPaymentCalendar	TGT	TGT
FloatFirstStubDate		
FloatLastStubDate		
FloatRollDay	NORMAL	NORMAL
FloatFixingLag	0D	0D
FloatFirstFixing		
FloatLastFixing		
FloatPaymentLag	0D	0D
FloatStubType	ShortStart	ShortStart
FloatSpread	13.90	0.00

Swap Pricing & Risk

	Leg1	Leg2	Total
Swap Par Rate	0.0161%	0.0161%	0.0000%
Swap PV	-814	814	0
Swap PV01	505	-505	0
Swap DV01	503	-503	0

3) Main 4) Details 5) Curves 6) Cashflow 7) Resets 9) Scenario 10) Risk 11)			
Deal		EUR Basis Swaps Counterparty	
Swap		SWAP CNTRPARTY	
Leg 1:Float		3 Month Euribor	
Notional	Receive 1MM	Notional	Pay 1MM
Currency	EUR	Currency	EUR
Effective	0D 06/16/2016	Effective	0D 06/16/2016
Maturity	5Y 06/16/2021	Maturity	5Y 06/16/2021
Index	6M EUR006M	Index	3M EUR003M
Spread	0.000 bp	Spread	14.822 bp
Leverage	1.00000	Leverage	1.00000
Latest Index	-0.15900	Latest Index	-0.26200
Reset Freq	SemiAnnual	Reset Freq	Quarterly
Pay Freq	SemiAnnual	Pay Freq	SemiAnnual
Day Count	ACT/360	Day Count	ACT/360
Market			
Leg 1: NPV	-1,705.09	Leg 2: NPV	1,705.09
Accrued	0.00	Accrued	0.00
Premium	-0.17	Premium	0.17
DV01	-506.49	DV01	506.77
Valuation Results			
Principal	0.00	Premium	0.00000
Accrued	0.00	BP Value	0.00000
NPV	0.00		

Asset Swap Spreads

meProductAssetSwapSpread(DealInfoLVB)

Bond Description

Bond	DBR 0.5% 15/02/2026
BondCleanPrice	103.950
FixedRate	0.5000%
MaturityDate	Sun 15-Feb-26

Swap Deal Info

Bond	DBR 0.5% 15/02/2026
BondCleanPrice	103.950
Notional	1,000,000
PayerReceiver	RECEIVER
EffectiveDate	Thu 2-Jun-16
MaturityDate	Sun 15-Feb-26
CurveCollection	EURYC
ForecastCurve	EUR6ML
DiscountCurve	OIS
FixedFrequency	Annual
FixedDaycount	ACT/ACT
FixedAccrualBusinessDayAdjustment	MOD_FOLLOWING
FixedAccrualCalendar	TGT
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING
FixedPaymentCalendar	TGT
FixedRollDay	15
FixedPaymentLag	OD
FixedStubType	ShortStart
FixedRate	0.5000%
FloatFrequency	Semi-Annual
FloatDaycount	ACT/360
FloatFixingBusinessDayAdjustment	MOD_FOLLOWING
FloatFixingCalendar	TGT
FloatAccrualBusinessDayAdjustment	MOD_FOLLOWING
FloatAccrualCalendar	TGT
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING
FloatPaymentCalendar	TGT
FloatFirstStubDate	
FloatLastStubDate	
FloatRollDay	15
FloatFixingLag	OD
FloatFirstFixing	
FloatLastFixing	
FloatPaymentLag	OD
FloatStubType	ShortStart
FloatSpread	-43.50

<--- Asset Swap Spread

Swap Pricing & Risk

Asset Swap Spread (bps)	-43.50
Swap PV	39,500
Par/Par Adjustment	39,500

DBR 0 ½ 02/15/26 €↑104.903 +.296 104.895/104.910 -.006 /-.008			
At 13:05 -- X -- Source BGN			
DBR 0 ½ 02/15/26 Actions Settings Asset Swap Calculator			
Pricing Cashflow Relative Value Deal Summary			
Asset Swap Analysis		Price	104.9000
Calculate		Z-Spread	-43.5
Price -> ASW Spread		Yield(%)	-0.00671
		ASW Spread	-43.1
		MMS Spread	-43.7
Bond JV503423 Swap Par-Par Matched Maturity 8) Swap Detail SWPM »			
Par Amount	1MM	Leg 1: Fixed	Pay
Workout	02/15/2026	Notional	1MM
Workout Price	100.0000	Currency	EUR
Pay Freq	Annual	Effective Date	01/15/2016
Day Count	ACT/ACT	Maturity Date	02/15/2026
		Coupon	0.5
		Pay/Reset Freq	Annual
		Day Count	ACT/ACT
Implied Value	100.6436	Include Accrued	
Market		Discount Curve	133 Mid
Curve Date	06/14/2016	Forward Curve	45 Mid
Settle Date	06/16/2016		
Swapped Spread Detail			
Clean Price	104.9000	Money	Spread(bp)
Swap Price	100.0000	Cash Out	-49,000.0
Swap Rate(%)	0.43389	Bond Cpn(%)	6,436.4
Redemption(%)	0.0000		0.0
Funding	Spread(bp)		0.0
Swapped Spread			-42,563.6

Swap Generators

Notional (millions)	1.00	1.00	1.00	1.00	1.00
Pay/Receive Fixed	RECEIVER	RECEIVER	RECEIVER	RECEIVER	RECEIVER
Fixed Rate	1.00%	1.00%	1.00%	1.00%	1.00%
Maturity	1Y	2Y	3Y	4Y	5Y
Float Spread	0.00	0.00	0.00	0.00	0.00
CSA	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA

OIS Deal Info

Notional	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
PayerReceiver	RECEIVER	RECEIVER	RECEIVER	RECEIVER	RECEIVER
EffectiveDate	02-Jun-16	02-Jun-16	02-Jun-16	02-Jun-16	02-Jun-16
CurveCollection	EURYC	EURYC	EURYC	EURYC	EURYC
Interpolation	Spline	Spline	Spline	Spline	Spline
ForecastCurve	OIS	OIS	OIS	OIS	OIS
DiscountCurve	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA	EURDF_NOCSA
CompoundMethod	GEOMETRIC	GEOMETRIC	GEOMETRIC	GEOMETRIC	GEOMETRIC
MaturityDate	1Y	2Y	3Y	4Y	5Y
FixedFrequency	Annual	Annual	Annual	Annual	Annual
FixedDaycount	Act/360	Act/360	Act/360	Act/360	Act/360
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING
FixedPaymentCalendar	TGT	TGT	TGT	TGT	TGT
FixedRollDay	NORMAL	NORMAL	NORMAL	NORMAL	NORMAL
FixedPaymentLag	0D	0D	0D	0D	0D
FixedStubType	ShortEnd	ShortEnd	ShortEnd	ShortEnd	ShortEnd
FixedRate	1.00%	1.00%	1.00%	1.00%	1.00%
FloatFrequency	Annual	Annual	Annual	Annual	Annual
FloatDaycount	Act/360	Act/360	Act/360	Act/360	Act/360
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING
FloatPaymentCalendar	TGT	TGT	TGT	TGT	TGT
FloatRollDay	NORMAL	NORMAL	NORMAL	NORMAL	NORMAL
FloatPaymentLag	0D	0D	0D	0D	0D
FloatStubType	ShortEnd	ShortEnd	ShortEnd	ShortEnd	ShortEnd
FloatSpread	0.00	0.00	0.00	0.00	0.00
FloatFixingLag	2D	2D	2D	2D	2D

OIS Pricing

PV	14,066.69	28,659.94	42,500.13	55,000.55	65,170.70
Par Rate	-0.3820%	-0.4010%	-0.3849%	-0.3425%	-0.2716%

Swap Generators

Notional (millions)	1.00	1.00	1.00	1.00	1.00
Pay/Receive Fixed	RECEIVER	RECEIVER	RECEIVER	RECEIVER	RECEIVER
Fixed Rate	1.00%	1.00%	1.00%	1.00%	1.00%
Maturity	1Y	2Y	3Y	4Y	5Y
FloatIndex	EUR3ML	EUR3ML	EUR3ML	EUR3ML	EUR3ML
Float Spread	0.00	0.00	0.00	0.00	0.00
CSA	EUROIS	EUROIS	EUROIS	EUROIS	EUROIS

Swap Deal Info

TradeID	IRS_1Y	IRS_2Y	IRS_3Y	IRS_4Y	IRS_5Y
TradeType	VanillaIRS	VanillaIRS	VanillaIRS	VanillaIRS	VanillaIRS
Notional	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
PayerReceiver	RECEIVER	RECEIVER	RECEIVER	RECEIVER	RECEIVER
EffectiveDate	Thu 2-Jun-16	Thu 2-Jun-16	Thu 2-Jun-16	Thu 2-Jun-16	Thu 2-Jun-16
MaturityDate	1Y	2Y	3Y	4Y	5Y
FixedFrequency	Annual	Annual	Annual	Annual	Annual
FixedDaycount	30/360	30/360	30/360	30/360	30/360
FixedPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING
FixedPaymentCalendar	TGT	TGT	TGT	TGT	TGT
FixedRollDay	NORMAL	NORMAL	NORMAL	NORMAL	NORMAL
FixedPaymentLag	0D	0D	0D	0D	0D
FixedStubType	ShortStart	ShortStart	ShortStart	ShortStart	ShortStart
FixedRate	1.00%	1.00%	1.00%	1.00%	1.00%
ForecastCurve	EUR3ML	EUR3ML	EUR3ML	EUR3ML	EUR3ML
FloatSpread	0.00	0.00	0.00	0.00	0.00
FloatFrequency	Quarterly	Quarterly	Quarterly	Quarterly	Quarterly
FloatDaycount	ACT/360	ACT/360	ACT/360	ACT/360	ACT/360
FloatPaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING	MOD_FOLLOWING
FloatPaymentCalendar	TGT	TGT	TGT	TGT	TGT
FloatRollDay	NORMAL	NORMAL	NORMAL	NORMAL	NORMAL
FloatFixingLag	0D	0D	0D	0D	0D
FloatPaymentLag	0D	0D	0D	0D	0D
FloatStubType	ShortStart	ShortStart	ShortStart	ShortStart	ShortStart
CurveCollection	EURYC	EURYC	EURYC	EURYC	EURYC
DiscountCurve	EUROIS	EUROIS	EUROIS	EUROIS	EUROIS

Swap Pricing & Risk

Swap Par Rate	-0.2859%	-0.2963%	-0.2650%	-0.2050%	-0.1247%
Swap PV	12,909	26,156	38,288	48,658	56,822
Swap PV01	100	202	303	404	505
Swap DV01	102	206	311	417	524

`meProductSwapDeltaLadder("TradeLabelValueBlock","CurveCollection","ForecastCurveSet","BumpSize","BumpMode")`

Risk Total by Trade	-103	-208	-307	-408	-509	-1,005	-1,899	-
Pillars	IRS_1Y	IRS_2Y	IRS_3Y	IRS_4Y	IRS_5Y	IRS_10Y	IRS_20Y	Total
EURYC_STD_O_N	0	0	0	0	0	0	0	0
EURYC_STD_T_N	0	0	0	0	0	0	0	0
EURYC_STD_LIBOR_1M	0	0	0	0	0	0	0	0
EURYC_STD_LIBOR_3M	-30	-30	0	0	0	0	0	-59
EURYC_STD_LIBOR_6M	0	0	0	0	0	0	0	0
EURYC_STD_LIBOR_12M	0	0	0	0	0	0	0	0
EURYC_STD_FUTURE_ERU6 COMDTY	-31	-31	0	0	0	0	0	-62
EURYC_STD_FUTURE_ERV6 COMDTY	7	6	0	0	0	0	0	13
EURYC_STD_FUTURE_ERX6 COMDTY	-9	-5	0	0	0	0	0	-14
EURYC_STD_FUTURE_ERZ6 COMDTY	-18	-21	0	0	0	0	0	-39
EURYC_STD_FUTURE_ERH7 COMDTY	-22	-28	0	0	0	0	0	-50
EURYC_STD_FUTURE_ERM7 COMDTY	0	-25	0	0	0	0	0	-25
EURYC_STD_FUTURE_ERU7 COMDTY	0	-25	0	0	0	0	0	-25
EURYC_STD_FUTURE_ERZ7 COMDTY	0	-27	0	0	0	0	0	-27
EURYC_STD_FUTURE_ERH8 COMDTY	0	-21	0	0	0	0	0	-21
EURYC_STD_3MLIBORSWAP_3Y	0	3	-303	0	0	0	0	-299
EURYC_STD_3MLIBORSWAP_4Y	0	-1	0	-404	0	0	0	-405
EURYC_STD_3MLIBORSWAP_5Y	0	0	0	0	-505	0	0	-505
EURYC_STD_3MLIBORSWAP_6Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_7Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_8Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_9Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_10Y	0	0	0	0	0	-1,002	0	-1,002
EURYC_STD_3MLIBORSWAP_11Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_12Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_15Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_20Y	0	0	0	0	0	0	-1,894	-1,894
EURYC_STD_3MLIBORSWAP_25Y	0	0	0	0	0	0	0	0
EURYC_STD_3MLIBORSWAP_30Y	0	0	0	0	0	0	0	0

Swap Delta Risk II

Risk Total by Trade	-103	-208	-307	-408	-509	-1,005	-1,899	-
Pillars	IRS_1Y	IRS_2Y	IRS_3Y	IRS_4Y	IRS_5Y	IRS_10Y	IRS_20Y	Total
EURYC_OIS_OUTRIGHTOISSWAP_ON	0	0	0	0	0	0	0	0
EURYC_OIS_CBRate_EUSF1A	0	0	0	0	0	0	0	0
EURYC_OIS_CBRate_EUSF2A	0	0	0	0	0	0	0	0
EURYC_OIS_CBRate_EUSF3A	0	0	0	0	0	0	0	0
EURYC_OIS_CBRate_EUSF4A	0	0	0	0	0	0	0	0
EURYC_OIS_OUTRIGHTOISSWAP_12M	-1	-1	-1	-1	-1	-1	-1	-8
EURYC_OIS_OUTRIGHTOISSWAP_18M	0	0	0	0	0	0	0	-1
EURYC_OIS_OUTRIGHTOISSWAP_2Y	0	-2	-3	-3	-3	-3	-3	-15
EURYC_OIS_BASISRATE_3Y	0	0	3	3	4	4	4	18
EURYC_OIS_3MLIBORSWAP_3Y	0	0	-3	-3	-4	-3	-4	-17
EURYC_OIS_BASISRATE_4Y	0	0	0	4	4	4	4	16
EURYC_OIS_3MLIBORSWAP_4Y	0	0	0	-4	-4	-4	-4	-16
EURYC_OIS_BASISRATE_5Y	0	0	0	0	5	4	4	12
EURYC_OIS_3MLIBORSWAP_5Y	0	0	0	0	-4	-4	-4	-12
EURYC_OIS_BASISRATE_6Y	0	0	0	0	0	3	3	6
EURYC_OIS_3MLIBORSWAP_6Y	0	0	0	0	0	-3	-3	-6
EURYC_OIS_BASISRATE_7Y	0	0	0	0	0	0	1	1
EURYC_OIS_3MLIBORSWAP_7Y	0	0	0	0	0	0	-1	-1
EURYC_OIS_BASISRATE_8Y	0	0	0	0	0	0	0	-1
EURYC_OIS_3MLIBORSWAP_8Y	0	0	0	0	0	0	0	1
EURYC_OIS_BASISRATE_9Y	0	0	0	0	0	-6	-5	-10
EURYC_OIS_3MLIBORSWAP_9Y	0	0	0	0	0	6	5	10
EURYC_OIS_BASISRATE_10Y	0	0	0	0	0	2	-1	1
EURYC_OIS_3MLIBORSWAP_10Y	0	0	0	0	0	-2	1	-1

Swap Pricer

Launcher: X:/Secure/DIR_MLib_Analytics/Pricing Tools/LaunchPricer.bat

Features

- Flexible Choice of Calibration Instruments & Broker Feeds
- Live Steamable Bloomberg Data
- Supports EUR, USD, GBP & JPY
- Flexible choice of CSA
- Swap Pricing, Strategies & Risk
- Includes Asset Swap Spread Solver

Swap Pricer: Settings

Valuation Date Thu, 16-Jun-16

Bloomberg Access TRUE

Pulse (Seconds) 15

Calc Mode MANUAL

CURVE BUILDING

	Curve Enablement	Native CSA	CSA Curves
EUR	TRUE	Success	Success
USD	TRUE	Success	Success
GBP	TRUE	Success	Success
JPY	TRUE	Success	Success

MARKET DATA

OIS CURVE Ready

	OIS SWAPS	OIS BASIS	LIBOR SWAPS
EUR	24	17	17
USD	20	12	12
GBP	20	0	0
JPY	34	0	0

SWAP CURVES Ready

	MM	FRAX3	FRAX6	FUT	SWAP
EUR	1	0	11	0	17
USD	1	0	0	12	10
GBP	1	0	8	0	17
JPY	1	0	6	0	17

BASIS CURVES Ready

	TENOR1	TENOR2	TENOR3	XCCY
EUR	16	14	16	15
USD	15	15	12	-
GBP	17	17	17	15
JPY	17	16	-	15

NAVIGATION

Show / Hide Calibration Instruments

Show / Hide Market Data

Show / Hide Curve Configurations

Show / Hide Curve Build & Charts

SWAP PRICING

Show / Hide Swap Pricing Screens

Show / Hide Custom Swap Screens

Show / Hide Asset Swap Screens

DATA STREAMING

ENABLE Data Streaming

CALIBRATION

Step 1 Request Market Data

Step 2 Update Calibration Parameters

Step 3 Build Curves

Swap Pricer: Vanilla Interest Rate Swaps

EUR Swaps 

PV 516,434

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First Fixing	Fixed Roll Day	Float Roll Day	CSA	Par Rate	Present Value	PV01	DV01
1	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	1Y	1.00000%	6ML	0.00				NOCSA	-0.14854%	11,482	100	101
2	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	2Y	1.00000%	6ML	0.00				NOCSA	-0.14700%	22,930	200	203
3	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	3Y	1.00000%	6ML	0.00				NOCSA	-0.11400%	33,401	300	307
4	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	4Y	1.00000%	6ML	0.00				NOCSA	-0.05600%	42,268	400	411
5	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	5Y	1.00000%	6ML	0.00				NOCSA	0.02500%	48,735	500	516
6	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	6Y	1.00000%	6ML	0.00				NOCSA	0.12200%	52,627	599	620
7	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	7Y	1.00000%	6ML	0.00				NOCSA	0.23000%	53,838	699	724
8	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	8Y	1.00000%	6ML	0.00				NOCSA	0.34400%	52,384	799	826
9	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	9Y	1.00000%	6ML	0.00				NOCSA	0.45500%	48,871	897	925
10	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	10Y	1.00000%	6ML	0.00				NOCSA	0.55600%	44,140	994	1,023
11	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	11Y	1.00000%	6ML	0.00				NOCSA	0.64800%	38,346	1,089	1,116
12	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	12Y	1.00000%	6ML	0.00				NOCSA	0.73200%	31,708	1,183	1,207
13	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	13Y	1.00000%	6ML	0.00				NOCSA	0.80476%	24,906	1,276	1,296
14	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	14Y	1.00000%	6ML	0.00				NOCSA	0.86687%	18,195	1,367	1,383
15	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	15Y	1.00000%	6ML	0.00				NOCSA	0.91900%	11,796	1,456	1,467
16	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	16Y	1.00000%	6ML	0.00				NOCSA	0.96220%	5,839	1,545	1,550
17	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	17Y	1.00000%	6ML	0.00				NOCSA	0.99708%	477	1,631	1,632
18	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	18Y	1.00000%	6ML	0.00				NOCSA	1.02553%	-4,384	1,717	1,712
19	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	19Y	1.00000%	6ML	0.00				NOCSA	1.04824%	-8,688	1,801	1,791
20	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	20Y	1.00000%	6ML	0.00				NOCSA	1.06600%	-12,436	1,884	1,870

Swap Pricer: Customized Interest Rate Swaps

EUR Swaps 

Float Index

IMM, EOM Rolls

CSA

PV

513,542

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First Fixing	Fixed Roll Day	Float Roll Day	CSA	Par Rate	Present Value	PV01	DV01
1	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	1Y	1.000000%	6ML	0.00		IMM	IMM	NOCSA	-0.14857%	11,482	100	101
2	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	2Y	1.000000%	6ML	0.00		IMM	IMM	NOCSA	-0.14701%	22,930	200	203
3	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	3Y	1.000000%	6ML	0.00		IMM	IMM	NOCSA	-0.11401%	33,401	300	307
4	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	4Y	1.000000%	6ML	0.00		IMM	IMM	NOCSA	-0.05603%	42,269	400	411
5	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	5Y	1.000000%	3ML	0.00				NOCSA	0.02500%	48,735	500	516
6	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	6Y	1.000000%	3ML	0.00				NOCSA	0.12200%	52,627	599	620
7	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	7Y	1.000000%	6ML	0.00		EOM	EOM	NOCSA	0.23022%	53,823	699	724
8	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	8Y	1.000000%	6ML	0.00		EOM	EOM	NOCSA	0.34416%	52,372	799	826
9	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	9Y	1.000000%	6ML	0.00		EOM	EOM	NOCSA	0.45506%	48,867	897	925
10	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	10Y	1.000000%	6ML	0.00		EOM	EOM	NOCSA	0.55606%	44,135	994	1,023
11	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	11Y	1.000000%	6ML	0.00				USDCSA	0.64800%	38,346	1,089	1,116
12	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	12Y	1.000000%	6ML	0.00				USDCSA	0.74314%	31,849	1,240	1,264
13	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	13Y	1.000000%	6ML	0.00				USDCSA	0.81726%	24,493	1,340	1,361
14	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	14Y	1.000000%	6ML	0.00				EURCSA	0.88054%	17,198	1,440	1,455
15	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	15Y	1.000000%	6ML	0.00				EURCSA	0.93364%	10,206	1,538	1,547
16	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	16Y	1.000000%	6ML	0.00				GBPCSA	0.96220%	5,839	1,545	1,550
17	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	17Y	1.000000%	6ML	0.00				GBPCSA	0.99708%	477	1,631	1,632
18	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	18Y	1.000000%	6ML	0.00				JPYCSA	1.02553%	-4,384	1,717	1,712
19	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	19Y	1.000000%	6ML	0.00				JPYCSA	1.04824%	-8,688	1,801	1,791
20	Mon 20-Jun-16	2D	1,000,000	EUR	RECEIVER	20Y	1.000000%	6ML	0.00				JPYCSA	1.06600%	-12,436	1,884	1,870

Swap Pricer: Asset Swap Spreads

Bond ISIN

DE0001102374 Govt



1

Request Bond Static Data

Bloomberg Bond Data

Ticker	Security Name	Maturity Date	Coupon	Last Price	Daycount	Coupon Freq	Coupon Desc
DE0001102374 Govt	DBR 0 1/2 02/15/25	15-Feb-25	0.50	105.6150	ACT/ACT	1	Annual

EUR_ASSET_SWAP

	BOND LEG	FLOAT LEG
Effective Date	Mon 20-Jun-16	2D
MaturityOrTenor	Sat 15-Feb-25	
Yield Curve ID	EURYC	
Rate / Spread	0.50000%	0.00
frequency	Annual	Semi-Annual
daycount	ACT/ACT	ACT/360
rollConv	MOD_FOLLOWING	MOD_FOLLOWING
calendar	TGT	TGT
firstStub		
lastStub		
rollDay	15	15
fixingLag	-	0D
firstFixing	-	-0.14300%
lastFixing	-	
paymentLag	0D	0D
StubType	ShortStart	ShortStart
Interpolation	Spline	
Forecast Curve	EUR6ML	
Discount Curve	EURDF_NOCSA	NOCSA
Interp Forwards	TRUE	
EOM Roll	TRUE	

mirAssetsSwapSpread

-56.01



3

Refresh Quote

2

Par-Par Asset Swap Spread

Bloomberg BPIPE Request

Requested EUR_ASSET_SWAP @ 11:20:19

Security Name	SECURITY_DESCRIPTION_RT
Maturity Date	MATURITY_RT
Coupon	COUPON_RT
Last Price	LAST_PRICE
Daycount	DAY_COUNT_DESCRIPTION_REALTIME
Coupon Freq	CPN_FREQ
Coupon Desc	COUPON_FREQUENCY_DESCRIPTION

Documentation Folder:

[fs002/Global/Secure/DIR_MLib_Analytics/Documentation](#)

- **Ametrano & Bianchetti (2013)**
Everything You Always Wanted to Know About Multiple Interest Rate Curve Bootstrapping But Were Afraid to Ask
- **Banca IMI, Ametrano (2011)**
Rate Curves for Forward Euribor Estimation and CSA Discounting
- **Den Danske Bank, Christian Fenger (2007)**
Super Surface
- **Mizuho, Nicholas Burgess (2016)**
Yield Curve Overview: A Guide on How to Build CSA Curves with Excel Example(s)

The End